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International Institute of Information Technology, Naya Raipur

(A Joint Initiative of NTPC Ltd. & Government of Chhattisgarh)

Atal Nagar – Naya Raipur, Chhattisgarh – 493661

SYLLABUS BOOK

Bachelor of Technology (B.Tech.)

Electronics and Communication Engineering (ECE)

Curriculum & Detailed Syllabi

Batch 2025 Onwards • Total Credits: 160

(With effect from July 2025)

Designed under the Outcome-Based Education (OBE) framework
aligned with NBA & NEP 2020

Department of Electronics & Communication Engineering

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1. About IIIT-NR

Dr. Shyama Prasad Mukherjee International Institute of Information Technology, Naya Raipur (IIIT-NR) is a premier state-funded institute established in 2015 under the International Institute of Information Technology University Act, 2013 of the Government of Chhattisgarh. The institute is a joint initiative of the Government of Chhattisgarh and National Thermal Power Corporation (NTPC) Limited, a Maharatna Central Public Sector Undertaking under the Ministry of Power, Government of India.

Incorporated as a state university and recognized by the University Grants Commission (UGC) and approved by the All India Council for Technical Education (AICTE), IIIT-NR is committed to pursuing excellence in higher education, research, and development in Information Technology (IT) and associated disciplines. The institute firmly believes in knowledge dissemination and in imparting entrepreneurial skills to its students, enabling them to address real-world problems with confidence and creativity.

The institute is housed on a fully residential, fifty-acre green campus in Naya Raipur (Atal Nagar) — India's first planned green and smart city — approximately 23 km from Raipur and well connected by air, rail, and road. The serene environment blends natural beauty with state-of-the-art infrastructure, creating an ideal space for innovation and learning.

IIIT-NR offers undergraduate (B.Tech.), postgraduate (M.Tech.), and doctoral (Ph.D.) programmes in Computer Science and Engineering (CSE), Electronics and Communication Engineering (ECE), and Data Science and Artificial Intelligence (DSAI). These credit-based programmes are designed to foster core competence together with innovation and entrepreneurship, with a dynamic curriculum structure that keeps pace with the changing needs of industry and academia.

Vision of the Institute

To be a globally recognized institution of excellence in information technology and allied disciplines, nurturing innovators, entrepreneurs, and socially responsible technologists who contribute to national and global development.

Mission of the Institute

- To impart high-quality, value-based, and outcome-oriented technical education.
- To foster state-of-the-art, interdisciplinary research and product development.
- To cultivate entrepreneurial mind-sets and industry-ready professional skills.
- To engage with society and industry for sustainable and inclusive growth.

Key Highlights

Particular	Details
Established	2015 (Act 15 of 2013, Government of Chhattisgarh)
Type	State University (Joint venture of Govt. of Chhattisgarh & NTPC Ltd.)
Recognition	UGC-recognized; AICTE-approved
Campus	50-acre fully residential green campus, Naya Raipur (Atal Nagar)
Programmes	B.Tech. (CSE / ECE / DSAI), M.Tech., Ph.D.
Motto	Knowledge gives us Intense Peace

2. About the Department of ECE

The Department of Electronics and Communication Engineering (ECE) at IIIT-NR offers a four-year Bachelor of Technology programme that provides a strong conceptual foundation in electronics, communication, signal processing, and VLSI, complemented by intensive hands-on experience in designing and building electronic hardware and embedded systems. The programme has been developed in response to the evolving needs of industry, equipping students with advanced knowledge and skills for the design of complex electronic and communication systems.

The department is taught by highly qualified and research-active faculty whose expertise spans wireless and next-generation communications (5G/6G, OTFS, UAV networks), VLSI and microelectronics, embedded and IoT systems, RF and microwave engineering, signal processing, control, and machine learning for communications. Every semester integrates project work that combines coding and electronics skills, supported by well-equipped laboratories and the active mentorship of research scholars.

Vision of the Department

To emerge as a centre of excellence in electronics and communication engineering education and research, producing globally competent engineers and innovators capable of addressing the technological challenges of a connected world.

Mission of the Department

1. To provide a rigorous, outcome-based foundation in electronics and communication engineering with strong hands-on and design exposure.
2. To promote research, innovation, and entrepreneurship in emerging areas such as next-generation wireless, VLSI, embedded systems, and intelligent electronics.
3. To inculcate professional ethics, lifelong learning, and social responsibility among students.
4. To build strong linkages with industry, research organizations, and the wider community for mutually beneficial collaboration.

Programme Offered

B.Tech. in Electronics and Communication Engineering — 4 years / 8 semesters, 160 credits, with multiple entry-exit options, multidisciplinary minors, an Honors track, and a strong project and internship component, in line with NEP 2020 and the NBA outcome-based education framework.

Credit Structure and Break-up

The programme follows the NEP-based B.Tech. Common Format credit scheme, with a total programme credit requirement of 160 credits distributed across Lecture (L), Tutorial (T), Practical (P), and Scholarly Engagement Hours (S) components as defined below:

L — Lecture	1 Credit = 1 hour / week
T — Tutorial	1 Credit = 1 hour / week
P — Practical	1 Credit = 2 hours / week
S — Scholarly Engagement Hours	0 Credit = 2 hours / week

Salient Features of the Curriculum

- Choice-based credit system with Program Electives (Communications, Embedded Systems, VLSI tracks), Open Electives, and Multidisciplinary Minors.

- AI-assisted full-stack prototyping, IoT, 3D printing, and FPGA-based design embedded from the first year.
- Strong project spine: Mini Project, Project-I & II, Minor & Major Projects, and a mandatory summer internship.
- Honors option (additional 15 credits) for high-performing students with a research publication / patent requirement.
- Value-added courses on Indian Knowledge Systems, ethics, constitution, finance, and community service for holistic development.

3. Outcome-Based Education (OBE) Framework

The B.Tech. (ECE) curriculum is designed and delivered following the Outcome-Based Education (OBE) philosophy advocated by the National Board of Accreditation (NBA) and consistent with the Washington Accord. Under OBE, the curriculum, teaching-learning processes, and assessment are aligned to clearly defined outcomes at three levels — Programme Educational Objectives (PEOs), Programme Outcomes (POs) with Programme Specific Outcomes (PSOs), and Course Outcomes (COs).

The OBE Hierarchy

Outcome Level	Description & Time-frame
PEOs	Broad statements describing the career and professional accomplishments graduates are expected to attain within 3–5 years of graduation.
POs / PSOs	Statements of knowledge, skills, and attitudes students should possess at the time of graduation; PSOs are programme-specific competencies (ECE).
COs	Specific, measurable abilities a student acquires on completing an individual course; COs are mapped to POs/PSOs and assessed each semester.

Attainment is measured through direct assessment (continuous internal evaluation, end-semester examinations, laboratory and project rubrics) and indirect assessment (course-exit, programme-exit, alumni, and employer surveys). The closed feedback loop drives continuous quality improvement (CQI).

3.1 Programme Educational Objectives (PEOs) and Justification

Within a few years of graduation, B.Tech. (ECE) graduates of IIIT-NR will:

PEO	Statement	Justification
PEO-1	Establish themselves as competent professionals in electronics, communication, VLSI, embedded, and allied industries, or pursue higher studies and research in reputed institutions.	Reflects the strong technical core (electronics, signals, communication, VLSI) and project spine that build employability and research readiness.
PEO-2	Apply engineering knowledge and modern tools to analyse, design, and implement sustainable solutions to real-world and societal problems.	Supported by design-oriented courses, open-source EDA/simulation tools, IoT, and SDG/environment components in the curriculum.
PEO-3	Demonstrate effective communication, teamwork, leadership, ethical conduct, and entrepreneurial thinking in multidisciplinary environments.	Addressed by HSMC courses (language, discourse, ethics, finance), entrepreneurship & design thinking, and team-based projects.
PEO-4	Engage in lifelong learning and adapt to emerging technologies such as 5G/6G, AI/ML, quantum and intelligent electronic systems.	Enabled by foundation in AI/ML, emerging-technology electives, MOOC/NPTEL pathways, and research methodology training.

3.2 Programme Outcomes (POs)

On successful completion of the programme, graduates will be able to demonstrate the twelve NBA Graduate Attributes (PO-1 to PO-12):

PO	Attribute	Statement
PO-1	**Engineering knowledge**	Apply knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
PO-2	**Problem analysis**	Identify, formulate, review research literature, and analyse complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
PO-3	**Design/development of solutions**	Design solutions for complex engineering problems and design system components or processes that meet specified needs with appropriate consideration for public health and safety, and cultural, societal, and environmental considerations.
PO-4	**Investigation of complex problems**	Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of information to provide valid conclusions.
PO-5	**Modern tool usage**	Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modelling to complex engineering activities with an understanding of the limitations.
PO-6	**The engineer and society**	Apply reasoning informed by contextual knowledge to assess societal, health, safety, legal, and cultural issues and the consequent responsibilities relevant to professional engineering practice.
PO-7	**Environment and sustainability**	Understand the impact of professional engineering solutions in societal and environmental contexts, and demonstrate knowledge of and need for sustainable development.
PO-8	**Ethics**	Apply ethical principles and commit to professional ethics and responsibilities and norms of engineering practice.
PO-9	**Individual and team work**	Function effectively as an individual, and as a member or leader in diverse teams and in multidisciplinary settings.
PO-10	**Communication**	Communicate effectively on complex engineering activities with the engineering community and with society at large.
PO-11	**Project management and finance**	Demonstrate knowledge and understanding of engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
PO-12	**Life-long learning**	Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

3.3 Programme Specific Outcomes (PSOs)

In addition to the POs, ECE graduates will be able to:

PSO	Statement
PSO-1	Analyse, design, and simulate analog, digital, RF, and mixed-signal electronic circuits and VLSI systems using industry-standard and open-source EDA tools.
PSO-2	Model, design, and evaluate modern communication and signal-processing systems — including wireless, optical, and next-generation (5G/6G) networks — for real-world applications.

PSO	Statement
PSO-3	Develop embedded, IoT, and intelligent electronic systems by integrating hardware, firmware, and machine-learning techniques to deliver sustainable engineering solutions.

3.4 PEO – PO / PSO Mapping

The following matrix maps each PEO to the POs and PSOs it primarily contributes to (✓ = strong correlation).

PEO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
PEO-1	✓	✓	✓	✓	✓						✓	✓	✓	✓	✓
PEO-2	✓	✓	✓	✓	✓	✓	✓				✓		✓	✓	✓
PEO-3			✓			✓	✓	✓	✓	✓	✓				✓
PEO-4	✓	✓		✓	✓		✓					✓	✓	✓	✓

Course-level CO–PO/PSO mapping matrices are maintained by the department for every course and are reviewed each semester as part of the continuous quality improvement (CQI) process.

4. Industry Relevance of the Subjects

The ECE curriculum is intentionally aligned with the competencies sought by the electronics, semiconductor, telecommunication, embedded-systems, and IT industries, as well as with national missions such as the India Semiconductor Mission, Digital India, and Atmanirbhar Bharat. The table below maps subject clusters to their industry relevance.

Subject Cluster	Industry Relevance & Typical Roles
Mathematics & Computing core (Linear Algebra, Calculus, Probability & Statistics, Optimization, C, Data Structures, DAA, OOP)	Foundation for signal processing, communication, ML, EDA algorithms, and software-defined systems used across semiconductor, telecom, and IT industries.
Digital Electronics, FPGA & VLSI (Digital Electronics with Verilog, FPGA System Design, VLSI Design & IC Technology, Testing & Verification, Microelectronics)	Directly relevant to semiconductor design houses and India's Semiconductor Mission — RTL design, verification, physical design, and DFT roles at firms such as Intel, Qualcomm, NXP, Synopsys, Cadence, and Texas Instruments.
Signals, Systems & DSP (Signals and Systems, Digital Signal Processing, Control Systems)	Core to audio/image/biomedical processing, radar, automotive ADAS, and control of drones and robotics; widely used in product R&D and embedded DSP roles.
Communication systems (Communication System, Digital Communication I & II, Information & Coding)	Backbone of telecom and networking (5G/6G, satellite, optical) at Ericsson, Nokia, Reliance Jio, Qualcomm, and defence/space (ISRO, DRDO).
RF, Microwave & Antennas (Electromagnetic Field Theory, Applied Antenna Engineering, Microwave/mmWave & RFIC)	Essential for RF front-end, mmWave 5G, radar, and antenna design in wireless, aerospace, and defence sectors.
Embedded, IoT & Robotics (Internet of Things, Embedded Systems Design, Introduction to Robotics & UAV, Electronic Circuits)	Powers smart devices, industrial automation, automotive electronics, and UAV/robotics startups; high demand for embedded-firmware engineers.
AI/ML & Data (Introduction to AI & ML, Machine Learning Algorithms)	Enables AI-on-edge, intelligent communication, predictive maintenance, and data-driven design across virtually every modern electronics/telecom product.
Professional skills (Entrepreneurship & Design Thinking, IT Workshop, Finance, Communication, Ethics, IKS)	Build the workplace-readiness, innovation, and managerial competence valued by recruiters and essential for entrepreneurs and team leaders.

Course-level industry relevance is further detailed within each course syllabus in Section 6.

5. Detailed Scheme of Syllabi (Semesters 1–7)

Credit & contact-hour scheme of the B.Tech. (ECE) programme, Batch 2025 onwards. Total programme credits: 160.

Legend — L: Lecture (1 credit = 1 hr/week); T: Tutorial (1 credit = 1 hr/week); P: Practical (1 credit = 2 hr/week); S: Scholarly Engagement Hours (0 credit = 2 hr/week). Type — BSC: Basic Science; ESC: Engineering Science; PCC: Programme Core; PSC: Programme Specific Core; VSEC: Vocational/Skill Enhancement; HSMC: Humanities, Social Science & Mgmt; MDM: Multidisciplinary Minor; PEC: Programme Elective; OE/OEC: Open Elective; ELC/EL: Experiential Learning; CCA/CEA: Co-/Extra-curricular.

Semester 1 — Total Credits: 21

Sl.	Course Name	Code	Type	Cr.	L-T-P-S
1	Linear Algebra and Matrix Analysis	MA-L-01-C	BSC	3	2-1-0-0
2	Calculus	MA-L-02-C	BSC	3	2-1-0-0
3	Problem Solving with C Programming	CS-L-01-C	ESC/VSEC	3	2-0-2-2
4	Digital Electronics using Verilog	EC-L-01-C	ESC	3	2-0-2-0
5	International Language Competency	HS-L-01-C	HSMC (AEC)	2	1-0-2-2
6	IT Workshop: Full Stack Prototyping with AI Support	DA-L-01-C	VSEC	2	1-0-2-2
7	Entrepreneurship and Design Thinking	HS-L-02-C	HSMC	2	1-0-2-2
8	Internet of Things	EC-L-02-C	VSEC	2	1-0-2-2
9	Environmental Studies	HS-L-03-C	HSMC (VEC)	1	1-0-0-0
	Total			**21**	**14-2-10**

Semester 2 — Total Credits: 21

Sl.	Course Name	Code	Type	Cr.	L-T-P-S
1	Quantum Mechanics	PH-L-01-C	BSC	3	2-1-0-0
2	Probability and Statistics	MA-L-03-C	BSC	3	2-1-0-0
3	Data Structures	CS-L-02-C	ESC/VSEC	3	2-0-2-2
4	Signals and Systems	EC-L-03-C	ESC	3	2-0-2-0
5	Introduction to AI & ML	DA-L-02-C	ESC/VSEC	2	1-0-2-2
6	Communication and Discourse Strategies	HS-L-04-C	HSMC (AEC)	2	1-0-2-2
7	Human Values and Professional Ethics	HS-L-05-C	HSMC (VEC)	2	2-0-0-2
8	3D Printing and Prototype Design	EC-L-04-C	VSEC	2	1-0-2-2
9	Constitution of India	HS-L-06-C	HSMC (VEC)	1	1-0-0-0
	Total			**21**	**13-2-12**

Semester 3 — Total Credits: 22

Sl.	Course Name	Code	Type	Cr.	L-T-P-S
1	Optimization Techniques	MA-L-04-C	BSC	2	2-1-0-0
2	Digital Signal Processing	EC-L-05-C	PCC	3	2-0-2-2
3	Communication System	EC-L-06-C	PCC	3	2-0-2-2
4	Design and Analysis of Algorithms	CS-L-03-C	PCC	3	2-0-2-2
5	Object Oriented Programming	CS-L-04-C	PCC	2	1-0-2-2
6	Electronic Circuit-1	EC-L-07-C	PSC	3	1-0-2-0
7	Electromagnetic Field Theory	EC-L-08-C	PSC	2	1-0-2-0
8	Multi-Disciplinary Minor-1	MD-L-xx-E	MDM	3	3-0-0-0
9	Introduction to Finance	HS-L-07-C	HSMC (AEC)	1	2-0-0-0
	Total			**22**	**17-1-10**

Semester 4 — Total Credits: 25

Sl.	Course Name	Code	Type	Cr.	L-T-P-S
1	Machine Learning Algorithms	DA-L-03-C	PCC	3	2-0-2-2
2	Embedded Systems Design	EC-L-09-C	PCC	3	2-0-2-2
3	Digital Communication I	EC-L-10-C	PCC	3	2-0-2-2
4	Applied Antenna Engineering	EC-L-11-C	PCC	3	2-0-2-0
5	Electronics Circuit-2	EC-L-12-C	PCC	3	2-0-2-0
6	Network Analysis	EC-L-13-C	PCC	2	1-0-2-0
7	Image Processing and Computer Vision	EC-L-14-C	PCC	3	2-0-2-0
8	Multi-Disciplinary Minor-2	MD-L-xx-E	MDM	3	3-0-0-0
9	Experiential Learning through Mini Project	EC-L-15-C	EL	2	0-0-2-2
10	Community Services (NSS / ISR Activities)	HS-L-08-C	HSMC (CEA)	1	0-0-0-2
	Total			**25**	**15-0-12**

Semester 5 — Total Credits: 22

Sl.	Course Name	Code	Type	Cr.	L-T-P-S
1	Microelectronics	EC-L-16-C	PCC	3	2-0-2-2
2	Digital Communication II	EC-L-17-C	PCC	3	2-0-2-2
3	Control Systems	EC-L-18-C	PCC	3	2-0-2-0
4	VLSI Design and IC Technology	EC-L-19-C	PCC	3	2-0-2-2
5	FPGA based System Design	EC-L-20-C	PCC	2	1-0-2-0
6	Project - I (Research Methodology)	EC-L-21-C	ELC	2	0-0-4-2
7	Multi-Disciplinary Minor-3	MD-L-xx-E	MDM	3	3-0-0-0
8	Course on Indian Knowledge Systems	HS-L-09-C	HSMC (AEC)	2	1-0-2-0
	Total			**22**	**13-0-16**

Semester 6 — Total Credits: 25

Sl.	Course Name	Code	Type	Cr.	L-T-P-S
1	Microwave, mmWave & RFIC for Emerging Technologies	EC-L-22-C	PCC	3	2-0-2-2
2	Testing and Verification of VLSI Circuits	EC-L-23-C	PCC	2	1-0-2-0
3	Project - II	EC-L-24-C	ELC	2	0-0-4-0
4	Open Elective - I	OE-L-xx-E	OE	3	3-0-0-0
5	Program Elective - I	PE-L-xx-E	PEC	3	3-0-0-0
6	Program Elective - II	PE-L-xx-E	PEC	3	3-0-0-0
7	Program Elective - III	PE-L-xx-E	PEC	3	3-0-0-0
8	Multi-Disciplinary Minor-4	MD-L-xx-E	MDM	3	3-0-0-0
9	Indian/Foreign Language	HS-L-10-C	HSMC (AEC)	1	1-0-2-0
10	Co-Curricular & Extracurricular Activities	HS-L-11-C	CCA	2	-
	Total			**25**	**15-0-8**

Summer Semester — Internship (ELC, 3 credits): evaluated at the beginning of Semester 7; credit and grades reflected in the 7th-semester grade sheet.

Semester 7 — Total Credits: 19

Sl.	Course Name	Code	Type	Cr.	L-T-P-S
1	Minor Project	EC-L-25-C	ELC	2	0-0-4-0
2	Program Elective - IV	PE-L-xx-E	PEC	3	3-0-0-0
3	Program Elective - V	PE-L-xx-E	PEC	3	3-0-0-0
4	Open Elective - II	OE-L-xx-E	OEC	3	3-0-0-0
5	Multi-Disciplinary Minor-5	MD-L-xx-E	MDM	3	3-0-0-0
6	IKS Elective (History of Indian Civilizations / Kautilya's Arthashastra / Vedic Mathematics / Vedic Corpus / Panini's Grammar)	HS-L-12-C	HSMC (IKS)	1	2-0-0-0
7	Internship (Summer Semester)	EC-L-26-C	ELC	3	0-0-0-6
	Total			**18**	**14-0-4**

Semester 8 — Total Credits: 6

Sl.	Course Name	Code	Type	Cr.	L-T-P-S
1	Major Project (may be completed as Internship-cum-Project; courses may be joined via MOOC/NPTEL)	EC-L-27-C	ELC	6	-
	Total			**6**	**-**

Exit Options & Honors

- Exit after Year 1, Year 2, or Year 3 is permitted as per ordinance, subject to additional skill-based courses (6 credits) completed in summer, as per the list floated by the department.
- Honors: an additional credential earned through 15 extra credits in the discipline (minimum 8 CGPI, no backlog); students may opt for it after the 3rd semester. The 8th-semester project must be academic, with at least one Tier I/II conference or SCI-journal paper communicated, or one patent filed.

6. Detailed Course-wise Syllabus (Core Subjects)

This section presents the detailed Outcome-Based Education (OBE) syllabus for the core (compulsory) courses of the programme, covering the Basic Science (BSC), Engineering Science (ESC), Programme Core (PCC), and Programme Science (PSC) courses from Semester 1 to Semester 6. Each course is described using a uniform OBE template comprising: course information, Course Outcomes (COs), a five-module detailed syllabus with contact hours, text and reference books, industry relevance, Open Educational Resources (OERs) and open repositories, suggested laboratory exercises, and recommended open-source simulators/tools.

Professional (Programme) Electives and Open Electives are listed separately in Sections 7 and 8. Value-added, HSMC, and skill courses follow institute-wide common syllabi and are reflected in the scheme. Multidisciplinary Minor (MDM) streams are excluded as per the scope of this book.

MA-L-01-C Linear Algebra and Matrix Analysis

Course Code	MA-L-01-C	**Credits**	3
Category	BSC (Basic Science)	**Structure (L-T-P-S)**	2-1-0-0
Prerequisite	Nil	**Laboratory**	No

Provides the matrix-algebraic and vector-space foundation underpinning signal processing, communications, control, machine learning, and numerical computation in electronics engineering.

Course Outcomes (COs)

On successful completion of this course, the student will be able to:

CO	Statement
CO1	Solve systems of linear equations using Gaussian elimination and matrix inversion.
CO2	Carry out matrix operations including inverses, rank, and determinants.
CO3	Determine eigenvalues and eigenvectors and solve eigenvalue problems.
CO4	Apply matrix algebra to linear transformations and change of basis.
CO5	Demonstrate understanding of inner-product spaces, orthonormalization, and SVD.

Detailed Syllabus (5 Modules)

Mod.	Content	Hours
M1	**Systems of Linear Equations** System of linear equations; Gaussian elimination; inverse and rank of a matrix by elementary transformations; properties and applications of determinants.	5
M2	**Vector Spaces** Vector spaces over real and complex fields; subspaces; linear dependence and independence; spanning set; basis and dimension.	6
M3	**Linear Transformations** Linear transformations and their matrices; kernel and range; change of basis; rank-nullity theorem; matrix representation.	6
M4	**Eigenvalues & Diagonalization** Eigenvalues and eigenvectors; characteristic polynomial; diagonalization of matrices; Cayley-Hamilton theorem and applications.	6
M5	**Inner Product Spaces & SVD** Inner product spaces; norms; orthogonality; Gram-Schmidt orthonormalization; Singular Value Decomposition (SVD) and applications.	5
	Total Contact Hours (Lecture)	**28**

Text Books

1. S. Lipschutz, M. Lipson, Schaum's Outlines of Linear Algebra, McGraw-Hill.
2. G. Strang, Introduction to Linear Algebra, 4th ed., Wellesley-Cambridge Press, 2009.

Reference Books

1. J. Defranza, D. Gagliardi, Introduction to Linear Algebra with Applications, McGraw-Hill.
2. K. Hoffman, R. Kunze, Linear Algebra, Prentice Hall.

Industry Relevance

Linear algebra is the computational core of every modern signal-processing, communication (MIMO, beamforming), machine-learning, computer-vision, and EDA workload. SVD and eigen-analysis are routinely used in channel estimation, data compression, PCA, and recommender systems across the telecom, semiconductor, and AI industries.

Open Educational Resources (OERs) & Open Repositories

- MIT OCW 18.06 Linear Algebra (Gilbert Strang) — https://ocw.mit.edu/courses/18-06-linear-algebra-spring-2010/
- NPTEL: Linear Algebra — https://nptel.ac.in/courses/111106051
- 3Blue1Brown: Essence of Linear Algebra — https://www.3blue1brown.com/topics/linear-algebra
- Khan Academy: Linear Algebra — https://www.khanacademy.org/math/linear-algebra

Suggested Laboratory Exercises

1. Solve and visualize linear systems and row-reduction using Python/NumPy or GNU Octave.
2. Compute rank, inverse, and determinant for engineering matrices.
3. Eigen-decomposition and diagonalization of sample covariance matrices.
4. Image compression demonstration using SVD.

Open-Source Simulators / Tools

Python (NumPy, SciPy, Matplotlib) • GNU Octave • Sage • Jupyter

MA-L-02-C Calculus

Course Code	MA-L-02-C	**Credits**	3
Category	BSC (Basic Science)	**Structure (L-T-P-S)**	2-1-0-0
Prerequisite	Nil	**Laboratory**	No

Develops single- and multivariable calculus tools essential for modelling continuous systems, optimization, and field theory in engineering.

Course Outcomes (COs)

On successful completion of this course, the student will be able to:

CO	Statement
CO1	Find limits of functions graphically, numerically, and algebraically.
CO2	Apply continuity and differentiability to algebraic and transcendental functions.
CO3	Use partial derivatives, gradients, and Lagrange multipliers for optimization.
CO4	Apply the Fundamental Theorem of Calculus to evaluate definite integrals.
CO5	Evaluate multiple integrals to solve area, volume, and real-world problems.

Detailed Syllabus (5 Modules)

Mod.	Content	Hours
M1	**Single-Variable Calculus Review** Limit, continuity and differentiability; indeterminate forms and L'Hospital rule; Mean Value theorem; Maclaurin and Taylor series.	5
M2	**Functions of Several Variables** Limits and continuity; partial derivatives and differentiability; linearization and differentials; chain rule; gradient vector; tangent planes; directional derivatives.	6
M3	**Multivariable Optimization** Extreme values and saddle points; Lagrange multipliers; Taylor's formula for several variables; partial derivatives with constrained variables.	5
M4	**Double Integrals** Double integrals; change of order of integration; area and volume by double integral; double integrals in polar form.	6
M5	**Triple Integrals** Triple integrals in rectangular, cylindrical, and spherical coordinates; substitutions in multiple integrals; applications.	6
	Total Contact Hours (Lecture)	**28**

Text Books

1. M. D. Weir, J. Hass, Thomas' Calculus, 12th ed., Pearson.
2. G. B. Thomas, R. L. Finney, Calculus and Analytic Geometry, 9th ed., Pearson.

Reference Books

1. Hughes-Hallett et al., Calculus: Single and Multivariable, 6th ed., Wiley.
2. J. Stewart, Multivariable Calculus, Hybrid Edition, Cengage.

Industry Relevance

Differential and integral calculus underlie modelling of analog circuits, electromagnetic fields, control loops, and optimization in ML training (gradient descent). Multivariable optimization with constraints is a daily tool in design and resource-allocation problems in communication and VLSI.

Open Educational Resources (OERs) & Open Repositories

- MIT OCW 18.01 / 18.02 Calculus — https://ocw.mit.edu/courses/mathematics/
- NPTEL: Calculus of Several Real Variables — https://nptel.ac.in/courses/111105134
- Paul's Online Math Notes — https://tutorial.math.lamar.edu/
- Khan Academy: Multivariable Calculus — https://www.khanacademy.org/math/multivariable-calculus

Suggested Laboratory Exercises

1. Symbolic differentiation and integration with SymPy / Maxima.
2. Visualization of gradients, tangent planes, and saddle points.
3. Numerical evaluation of double/triple integrals.
4. Constrained optimization using Lagrange multipliers (case studies).

Open-Source Simulators / Tools

Python (SymPy, NumPy, Matplotlib) • Maxima • GeoGebra • Desmos

CS-L-01-C Problem Solving with C Programming

Course Code	CS-L-01-C	**Credits**	3
Category	ESC / VSEC	**Structure (L-T-P-S)**	2-0-2-2
Prerequisite	Nil	**Laboratory**	Mandatory (2 Hours)

Builds algorithmic problem-solving ability and procedural programming skills in C, the foundation for embedded and systems programming in electronics.

Course Outcomes (COs)

On successful completion of this course, the student will be able to:

CO	Statement
CO1	Identify and list the programming tasks involved in a computational problem.
CO2	Implement conditional branching, iteration, and recursion in C.
CO3	Use arrays, pointers, and structures to develop algorithms and programs.
CO4	Apply dynamic memory allocation and file handling in C.
CO5	Design modular C programs using functions and good coding practices.

Detailed Syllabus (5 Modules)

Mod.	Content	Hours
M1	**Fundamentals of C** Computer organization; computer languages; algorithm and flowchart; features and structure of a C program; basic elements: data types, operators, expressions, I/O.	5
M2	**Control Flow** Conditional instructions; iterations (for, while, do-while); unconditional statements (break, continue, goto); switch-case.	5
M3	**Arrays, Strings & Functions** One- and multi-dimensional arrays; strings; functions, parameter passing, scope; recursion.	6
M4	**Pointers** Pointers; pointers with arrays and functions; pointer arithmetic; dynamic memory allocation (malloc, calloc, realloc, free).	6
M5	**Structures, Unions & Files** Structures and unions; nested structures; arrays of structures; file handling: text and binary files, file operations.	6
	Total Contact Hours (Lecture)	**28**

Text Books

1. B. W. Kernighan, D. M. Ritchie, The C Programming Language, 2nd ed., Prentice Hall (ISBN 978-0131103627).
2. B. S. Gottfried, Schaum's Outline of Programming with C, McGraw-Hill (ISBN 978-0070240353).

Reference Books

1. E. Balagurusamy, Programming in ANSI C, McGraw-Hill.
2. Y. Kanetkar, Let Us C, BPB Publications.

Industry Relevance

C remains the dominant language for firmware, device drivers, RTOS, and microcontroller programming across the embedded, automotive, and IoT industries. Strong C and pointer skills are a baseline expectation for embedded-systems and systems-software roles.

Open Educational Resources (OERs) & Open Repositories

- NPTEL: Problem Solving through Programming in C — https://nptel.ac.in/courses/106105171
- CS50 Introduction to Computer Science (Harvard) — https://cs50.harvard.edu/x/
- GeeksforGeeks C Programming — https://www.geeksforgeeks.org/c-programming-language/
- learn-c.org Interactive Tutorial — https://www.learn-c.org/

Suggested Laboratory Exercises

1. Programs on operators, type conversion, and formatted I/O.
2. Branching and looping: prime check, factorial, patterns.
3. Array and string manipulation; searching and sorting.
4. Recursion: Fibonacci, Tower of Hanoi, GCD.
5. Pointers and dynamic memory; linked structures.
6. Structures, unions, and file read/write applications.

Open-Source Simulators / Tools

GCC / Clang • Code::Blocks • VS Code • Online GDB • Valgrind

EC-L-01-C Digital Electronics using Verilog

Course Code	EC-L-01-C	**Credits**	3
Category	ESC (Engineering Science)	**Structure (L-T-P-S)**	2-0-2-0
Prerequisite	Nil	**Laboratory**	Mandatory (2 Hours)

Introduces digital logic design and hardware description using Verilog HDL, forming the gateway to VLSI, FPGA, and embedded design.

Course Outcomes (COs)

On successful completion of this course, the student will be able to:

CO	Statement
CO1	Understand number systems, logic gates, Boolean simplification, and flip-flops.
CO2	Design basic digital systems using standard ICs and Verilog.
CO3	Analyse and design combinational circuits.
CO4	Analyse and design sequential circuits including counters and FSMs.
CO5	Implement and simulate digital designs in Verilog HDL.

Detailed Syllabus (5 Modules)

Mod.	Content	Hours
M1	**Number Systems & Boolean Algebra** K-maps; NAND/NOR implementations; tabular (Quine-McCluskey) simplification.	4
M2	**Combinational Design - Arithmetic** Design procedure; half/full adder and subtractor; 4-bit ripple-carry adder; carry-look-ahead adder; BCD arithmetic.	6
M3	**Combinational Design - MSI** Decoders, encoders, multiplexers, de-multiplexers, magnitude comparators; implementing logic with MUX/decoders.	6
M4	**Sequential Circuits** Latches and flip-flops; FF conversions; serial and parallel shift registers; synchronous and asynchronous counter design.	6
M5	**FSM & Verilog HDL** Mealy and Moore machines; state diagrams and tables; Verilog modelling styles (gate, dataflow, behavioral); test benches and simulation.	6
	Total Contact Hours (Lecture)	**28**

Text Books

1. R. P. Jain, Modern Digital Electronics, 3rd ed., Tata McGraw-Hill, 2003.
2. M. M. Mano, M. D. Ciletti, Digital Design with an Introduction to the Verilog HDL, 5th ed., Pearson, 2013.

Reference Books

1. T. L. Floyd, Digital Fundamentals, 10th ed., Pearson, 2011.
2. S. Palnitkar, Verilog HDL, 2nd ed., Pearson.

Industry Relevance

Digital design and Verilog are the entry skills for the semiconductor and FPGA industry. RTL design and simulation skills built here are directly used in chip-design houses (Intel, Qualcomm, NXP, Synopsys, Cadence) and align with the India Semiconductor Mission.

Open Educational Resources (OERs) & Open Repositories

- NPTEL: Digital Circuits — https://nptel.ac.in/courses/108105132
- NPTEL: Hardware Modeling using Verilog — https://nptel.ac.in/courses/106105165
- EDA Playground (online HDL simulator) — https://www.edaplayground.com/
- nandland.com Verilog & FPGA tutorials — https://nandland.com/

Suggested Laboratory Exercises

1. Verify logic gates and De Morgan's theorems in Verilog.
2. Design and simulate adders/subtractors and a 4-bit ALU.
3. Implement MUX, decoder, encoder, and comparator.
4. Design flip-flops and shift registers.
5. Design synchronous and asynchronous counters.
6. FSM design: sequence detector / traffic-light controller.

Open-Source Simulators / Tools

Icarus Verilog + GTKWave • Verilator • EDA Playground • Xilinx Vivado (free WebPACK) • Logisim Evolution

EC-L-02-C Introduction to Internet of Things

Course Code	EC-L-02-C	**Credits**	2
Category	VSEC	**Structure (L-T-P-S)**	1-0-2-2
Prerequisite	Nil	**Laboratory**	Mandatory

Introduces IoT architecture, sensors, microcontrollers, connectivity, and cloud platforms with hands-on prototyping.

Course Outcomes (COs)

On successful completion of this course, the student will be able to:

CO	Statement
CO1	Understand sensors, actuators, and signal conditioning in measurement systems.
CO2	Describe IoT architecture and real-world design constraints.
CO3	Interface sensors with Arduino / ESP32 / Raspberry Pi.
CO4	Use IoT communication protocols and cloud platforms.
CO5	Design and prototype IoT solutions for real applications.

Detailed Syllabus (5 Modules)

Mod.	Content	Hours
M1	**Introduction to IoT** IoT definition, history, and evolution; IoT vs traditional embedded systems; examples from smart cities, healthcare, agriculture, and industry.	3
M2	**IoT Architecture & Components** Layered architecture - perception, network, application; sensors, actuators, microcontrollers, communication modules.	3
M3	**Hardware Platforms** Arduino, ESP32/ESP8266, Raspberry Pi basics; development boards and their features; interfacing fundamentals.	3
M4	**Communication & Cloud** Wi-Fi, Bluetooth, Zigbee, LoRa, NB-IoT; choosing a protocol; MQTT; ThingSpeak, Blynk, Google/AWS IoT.	3
M5	**IoT Data & Security** Data collection, basic analytics and visualization; security challenges and good practices; mini-project launch.	2
	Total Contact Hours (Lecture)	**14**

Text Books

1. A. Bahga, V. Madisetti, Internet of Things: A Hands-On Approach, Universities Press.
2. J. Holler, V. Tsiatsis et al., From Machine-to-Machine to the Internet of Things, Elsevier.

Reference Books

1. R. Buyya, A. V. Dastjerdi, Internet of Things: Principles and Paradigms, Morgan Kaufmann.
2. Vendor documentation: ESP32, ThingSpeak, MQTT.

Industry Relevance

IoT skills feed directly into smart-manufacturing, automotive, agritech, and smart-city industries. Sensor interfacing, edge connectivity, and cloud dashboards are core to the rapidly growing IoT and Industry-4.0 job market.

Open Educational Resources (OERs) & Open Repositories

- NPTEL: Introduction to Internet of Things — https://nptel.ac.in/courses/106105166
- Wokwi IoT Simulator — https://wokwi.com/
- Tinkercad Circuits — https://www.tinkercad.com/circuits
- Hackster.io IoT Projects — https://www.hackster.io/projects

Suggested Laboratory Exercises

1. Blink and digital/analog I/O on Arduino/ESP32.
2. Read DHT11/LDR/PIR sensors and display values.
3. Log sensor data to ThingSpeak and visualize charts.
4. Relay/LED control via Blynk app (home automation).
5. Mini-project: smart plant watering / weather station.

Open-Source Simulators / Tools

Arduino IDE • Wokwi • Tinkercad Circuits • MQTT Explorer • ThingSpeak • Blynk

PH-L-01-C Quantum Mechanics

Course Code	PH-L-01-C	**Credits**	3
Category	BSC (Basic Science)	**Structure (L-T-P-S)**	2-1-0-0
Prerequisite	Nil	**Laboratory**	No

Introduces the principles and formalism of quantum mechanics that underpin modern electronics, photonics, semiconductor devices, and quantum technologies.

Course Outcomes (COs)

On successful completion of this course, the student will be able to:

CO	Statement
CO1	Use the language and formalism of basic quantum mechanics.
CO2	Carry out calculations for atoms and quantum phenomena in electron physics.
CO3	Interpret spectroscopic results in quantized units.
CO4	Solve the Schrödinger equation for standard potentials.
CO5	Relate quantum concepts to qubits and emerging quantum technologies.

Detailed Syllabus (5 Modules)

Mod.	Content	Hours
M1	**Foundations** Basic principles; wave function and uncertainty principle; probability amplitude and density; Schrödinger formalism; time-dependent and time-independent equations; eigenstates, eigenvalues, expectation values.	6
M2	**Wave Mechanics in 1-D** Schrödinger equation in one dimension; particle in a box; potential step and barrier; tunnelling; harmonic oscillator.	6
M3	**Angular Momentum & Central Potential** Orbital and spin angular-momentum operators; eigenstates/eigenvalues of angular momentum; central potential.	6
M4	**Hydrogen Atom** Solution of the Schrödinger equation in a central potential; hydrogen-like atom; quantum numbers and applications.	5
M5	**Dirac Formalism & Qubits** Dirac bra-ket algebra; angular momenta; Clebsch-Gordon coefficients; spin-orbit interaction; Hilbert space and the concept of qubits.	5
	Total Contact Hours (Lecture)	**28**

Text Books

1. A. Ghatak, S. Lokanathan, Quantum Mechanics: Theory and Applications.
2. D. J. Griffiths, Introduction to Quantum Mechanics, 2nd ed., Pearson.

Reference Books

1. L. I. Schiff, Quantum Mechanics, McGraw-Hill.
2. R. Shankar, Principles of Quantum Mechanics, Springer.

Industry Relevance

Quantum mechanics underlies semiconductor physics, lasers, photonics, and the emerging quantum-computing and quantum-communication industries. It is foundational for device-physics and R&D roles in semiconductor and quantum-technology organizations.

Open Educational Resources (OERs) & Open Repositories

- MIT OCW 8.04 Quantum Physics I — https://ocw.mit.edu/courses/8-04-quantum-physics-i-spring-2016/
- NPTEL: Quantum Mechanics — https://nptel.ac.in/courses/115106066
- IBM Quantum Learning — https://learning.quantum.ibm.com/
- Qiskit Textbook — https://qiskit.org/learn/

Suggested Laboratory Exercises

1. Numerical solution of the 1-D Schrödinger equation (Python).
2. Visualization of particle-in-a-box wavefunctions and probability densities.
3. Tunnelling probability vs barrier width simulation.
4. Single-qubit states and gates on the Bloch sphere using Qiskit.

Open-Source Simulators / Tools

Python (NumPy, QuTiP, Matplotlib) • Qiskit • IBM Quantum Composer

MA-L-03-C Probability and Statistics

Course Code	MA-L-03-C	**Credits**	3
Category	BSC (Basic Science)	**Structure (L-T-P-S)**	2-1-0-0
Prerequisite	Nil	**Laboratory**	No

Provides probability and statistical inference tools essential for communication theory, signal processing, reliability, and machine learning.

Course Outcomes (COs)

On successful completion of this course, the student will be able to:

CO	Statement
CO1	Apply probability rules including additive/multiplicative laws and Bayes' rule.
CO2	Analyse data using measures of central tendency, dispersion, and location.
CO3	Identify and use discrete and continuous distributions.
CO4	Translate real-world problems into probability models.
CO5	Perform estimation and hypothesis testing.

Detailed Syllabus (5 Modules)

Mod.	Content	Hours
M1	**Probability Foundations** Axiomatic definition; finite sample spaces; joint and conditional probability; independence; total probability; Bayes' rule and applications.	6
M2	**Random Variables** Discrete and continuous random variables; cdf, pmf, pdf and properties; bivariate and multivariate random variables; jointly distributed random variables; conditional and joint densities.	6
M3	**Distributions & CLT** Standard discrete and continuous distributions; sampling and sampling distributions; Central Limit Theorem.	6
M4	**Descriptive Statistics** Measures of central tendency, dispersion, skewness, kurtosis; data representation: histogram, pie chart, box-plot, bi-plot, multidimensional scaling.	5
M5	**Inference** Methods of estimation; unbiased estimators; confidence intervals (Z, t); hypothesis testing; types of errors; rejection-region and p-value approaches.	5
	Total Contact Hours (Lecture)	**28**

Text Books

1. A. J. Hayter, Probability and Statistics for Engineers, 4th ed., Cengage, 2012.
2. S. C. Gupta, V. K. Kapoor, Fundamentals of Mathematical Statistics, 11th ed., S. Chand, 2014.

Reference Books

1. S. M. Ross, Introduction to Probability and Statistics for Engineers and Scientists, 5th ed., Academic Press.

2. P. L. Meyer, Introductory Probability and Statistical Applications, Addison-Wesley.

Industry Relevance

Probability and statistics drive digital-communication theory (noise, detection, estimation), reliability engineering, A/B testing, and the entire machine-learning/data-science pipeline used across telecom, semiconductor, and analytics industries.

Open Educational Resources (OERs) & Open Repositories

- MIT OCW 6.041 Probabilistic Systems Analysis — https://ocw.mit.edu/courses/6-041-probabilistic-systems-analysis-and-applied-probability-fall-2010/
- NPTEL: Probability and Statistics — https://nptel.ac.in/courses/111105041
- Seeing Theory (visual probability) — https://seeing-theory.brown.edu/
- StatQuest (YouTube/Open) — https://statquest.org/

Suggested Laboratory Exercises

1. Simulate random experiments and verify probability laws (Python).
2. Fit and visualize standard distributions to data.
3. Demonstrate the Central Limit Theorem by sampling.
4. Confidence intervals and hypothesis tests on real datasets.

Open-Source Simulators / Tools

Python (NumPy, SciPy, statsmodels, seaborn) • R • Jamovi • Jupyter

CS-L-02-C Data Structures

Course Code	CS-L-02-C	**Credits**	3
Category	ESC / VSEC	**Structure (L-T-P-S)**	2-0-2-2
Prerequisite	Problem Solving with C Programming	**Laboratory**	Mandatory

Covers fundamental data structures and their applications, essential for efficient algorithm and embedded-software design.

Course Outcomes (COs)

On successful completion of this course, the student will be able to:

CO	Statement
CO1	Implement arrays and structures to solve real-time problems.
CO2	Apply stacks and queues to relevant problems.
CO3	Develop solutions using variations of stacks and queues.
CO4	Manage data using linked-list variations.
CO5	Devise tree- and graph-based application solutions.

Detailed Syllabus (5 Modules)

Mod.	Content	Hours
M1	**Stacks & Queues** Stacks; stacks using dynamic arrays; queues; circular queues; evaluation of expressions (infix/postfix).	6
M2	**Linked Lists** Singly, doubly, and circular linked lists; polynomials; additional list operations; applications.	6
M3	**Trees** Binary trees; tree traversals; additional binary-tree operations; heaps; binary search trees.	6
M4	**Graphs** Graph ADT; representations; elementary graph operations; BFS and DFS; spanning trees.	5
M5	**Searching, Sorting & Hashing** Linear and binary search; bubble, insertion, selection, quick, merge, heap sort; hashing and collision resolution.	5
	Total Contact Hours (Lecture)	**28**

Text Books

1. Horowitz, Sahni, Anderson-Freed, Fundamentals of Data Structures in C, 2nd ed., Universities Press, 2008.
2. Y. Langsam, M. Augenstein, A. Tannenbaum, Data Structures Using C and C++, 2nd ed., Pearson, 2003.

Reference Books

1. S. Lipschutz, G. A. V. Pai, Data Structures, Schaum's Outlines, McGraw-Hill, 2008.
2. R. F. Gilberg, B. A. Forouzan, Data Structures: A Pseudocode Approach with C, Cengage, 2005.

Industry Relevance

Efficient data structures are essential for embedded firmware, real-time systems, EDA tools, and any performance-critical software. Mastery of these is a universal screening criterion in technical interviews across the IT and core-electronics industries.

Open Educational Resources (OERs) & Open Repositories

- NPTEL: Data Structures and Algorithms — https://nptel.ac.in/courses/106102064
- VisuAlgo (algorithm visualization) — https://visualgo.net/
- GeeksforGeeks Data Structures — https://www.geeksforgeeks.org/data-structures/
- MIT OCW 6.006 Introduction to Algorithms — https://ocw.mit.edu/courses/6-006-introduction-to-algorithms-spring-2020/

Suggested Laboratory Exercises

1. Stack and queue implementation; expression evaluation.
2. Singly/doubly/circular linked-list operations.
3. Binary search tree and traversals.
4. Heap and priority-queue implementation.
5. Graph traversal (BFS/DFS) and shortest path.
6. Comparison of sorting algorithms and hashing.

Open-Source Simulators / Tools

GCC • VS Code • VisuAlgo • Python Tutor (visualization)

EC-L-03-C Signals and Systems

Course Code	EC-L-03-C	**Credits**	3
Category	ESC (Engineering Science)	**Structure (L-T-P-S)**	2-0-2-0
Prerequisite	Nil	**Laboratory**	Faculty Choice

Establishes the mathematical framework of continuous- and discrete-time signals and LTI systems, the bedrock of communication, DSP, and control.

Course Outcomes (COs)

On successful completion of this course, the student will be able to:

CO	Statement
CO1	Classify continuous and discrete signals by their properties.
CO2	Classify systems based on their properties.
CO3	Analyse continuous-time signals and systems with appropriate tools.
CO4	Analyse sampling and discrete-time signals.
CO5	Determine Fourier transforms for continuous- and discrete-time signals.

Detailed Syllabus (5 Modules)

Mod.	Content	Hours
M1	**Signals & Systems Basics** Classification of signals; signal operations (scaling, shifting, inversion); signal properties; elementary signals; classification and properties of systems.	6
M2	**LTI Systems & Convolution** Continuous-time LTI and discrete-time LSI systems; impulse and step response; convolution; system representation by differential/difference equations; eigenfunctions; frequency response.	6
M3	**Fourier Analysis** Fourier series of CT and DT signals; continuous-time Fourier transform and properties; Parseval's relation; time-bandwidth product.	6
M4	**DTFT, DFT & Sampling** Discrete-time Fourier transform and properties; relations among Fourier representations; DFT; sampling theorem; aliasing; reconstruction (ideal interpolator, ZOH, FOH).	5
M5	**Transform-Domain Analysis** Laplace/Z-transform: definition, region of convergence, properties; transform-domain analysis of LTI/LSI systems; system function, poles and zeros; stability.	5
	Total Contact Hours (Lecture)	**28**

Text Books

1. A. V. Oppenheim, A. S. Willsky, H. S. Nawab, Signals and Systems, Prentice Hall of India, 2006.
2. B. P. Lathi, Signal Processing and Linear Systems, Oxford University Press, 1998.

Reference Books

1. M. J. Roberts, Fundamentals of Signals and Systems, Tata McGraw-Hill, 2007.

2. R. F. Ziemer, W. H. Tranter, D. R. Fannin, Signals and Systems - Continuous and Discrete, 4th ed., Prentice Hall, 1998.

Industry Relevance

Signals and systems is the conceptual backbone of communication, audio/image/biomedical processing, radar, and control. The Fourier/Laplace/Z toolkit is used daily in DSP, telecom, and embedded-product R&D.

Open Educational Resources (OERs) & Open Repositories

- MIT OCW 6.003 Signals and Systems — https://ocw.mit.edu/courses/6-003-signals-and-systems-fall-2011/
- NPTEL: Signals and Systems — https://nptel.ac.in/courses/108104100
- Julius Smith - DSP-related Online Books (CCRMA) — https://ccrma.stanford.edu/~jos/
- Khan Academy / interactive Fourier demos — https://www.jezzamon.com/fourier/

Suggested Laboratory Exercises

1. Generate and operate on elementary signals in Python/Octave.
2. Compute and plot convolution of CT/DT signals.
3. Fourier series synthesis and CTFT/DTFT visualization.
4. DFT/FFT of signals and spectral analysis.
5. Sampling and aliasing demonstration; signal reconstruction.
6. Pole-zero plots and stability analysis.

Open-Source Simulators / Tools

Python (NumPy, SciPy, Matplotlib) • GNU Octave • Scilab

DA-L-02-C Introduction to AI & ML

Course Code	DA-L-02-C	**Credits**	2
Category	ESC / VSEC	**Structure (L-T-P-S)**	1-0-2-2
Prerequisite	Linear Algebra and Probability	**Laboratory**	Mandatory

Provides an intuitive, application-oriented introduction to AI and machine-learning workflows, tools, and core algorithms.

Course Outcomes (COs)

On successful completion of this course, the student will be able to:

CO	Statement
CO1	Distinguish AI from machine learning and deep learning.
CO2	Appraise the mathematics behind ML and the supervised/unsupervised paradigms.
CO3	Decide which ML paradigm suits a given real-world application.
CO4	Build and evaluate basic ML models using standard tools.
CO5	Recognize ethical and responsible-AI considerations.

Detailed Syllabus (5 Modules)

Mod.	Content	Hours
M1	**Introduction to AI & ML** History and evolution of AI; AI vs ML vs DL; real-world applications across sectors.	3
M2	**Data & Problem Formulation** Datasets, features, labels; types of ML problems - classification, regression, clustering; data preprocessing.	3
M3	**ML Workflow** Train/test split; model training; evaluation metrics - accuracy, precision, recall, F1; overfitting and validation.	3
M4	**Core Algorithms** Linear regression; decision trees; k-NN; Naïve Bayes; introduction to neural networks.	4
M5	**Tools & Responsible AI** Google Colab, Jupyter, Kaggle; bias, fairness, explainability, data privacy; competition launch.	1
	Total Contact Hours (Lecture)	**14**

Text Books

1. A. Géron, Hands-On Machine Learning with Scikit-Learn, Keras and TensorFlow, O'Reilly.
2. S. Raschka, Python Machine Learning, Packt Publishing.

Reference Books

1. T. Mitchell, Machine Learning, McGraw-Hill.
2. C. M. Bishop, Pattern Recognition and Machine Learning, Springer.

Industry Relevance

ML is now embedded in virtually every electronics and telecom product - from AI-on-edge and intelligent communication to predictive maintenance. Familiarity with the ML workflow and tooling is a strong differentiator for ECE graduates entering data-driven roles.

Open Educational Resources (OERs) & Open Repositories

- Google ML Crash Course — https://developers.google.com/machine-learning/crash-course
- NPTEL: Introduction to Machine Learning — https://nptel.ac.in/courses/106106139
- Kaggle Learn Micro-Courses — https://www.kaggle.com/learn
- scikit-learn User Guide — https://scikit-learn.org/stable/user_guide.html

Suggested Laboratory Exercises

1. Data exploration and preprocessing with pandas.
2. Linear/logistic regression on a sample dataset.
3. Decision tree and k-NN classification.
4. Model evaluation with metrics and cross-validation.
5. Kaggle beginner competition (e.g., Titanic).

Open-Source Simulators / Tools

Python (scikit-learn, pandas, matplotlib, seaborn) • Google Colab • Jupyter • Kaggle

MA-L-04-C Optimization Techniques

Course Code	MA-L-04-C	**Credits**	2
Category	BSC (Basic Science)	**Structure (L-T-P-S)**	2-1-0-0
Prerequisite	Linear Algebra, Calculus	**Laboratory**	Faculty Choice

Covers classical and modern optimization methods used in engineering design, resource allocation, communication, and machine learning.

Course Outcomes (COs)

On successful completion of this course, the student will be able to:

CO	Statement
CO1	Formulate engineering problems as optimization models.
CO2	Solve linear programming problems using simplex and duality.
CO3	Apply unconstrained and constrained nonlinear optimization methods.
CO4	Use gradient-based and numerical optimization techniques.
CO5	Apply optimization to communication, signal-processing, and ML problems.

Detailed Syllabus (5 Modules)

Mod.	Content	Hours
M1	**Introduction & LP** Classification of optimization problems; convex sets and functions; linear programming formulation; graphical method; simplex method.	6
M2	**Duality & Transportation** Duality theory; sensitivity analysis; transportation and assignment problems; network optimization.	6
M3	**Unconstrained Optimization** Single-variable methods; multivariable methods; gradient (steepest) descent; Newton and quasi-Newton methods; convergence.	6
M4	**Constrained Optimization** Lagrange multipliers; KKT conditions; penalty and barrier methods; quadratic programming.	5
M5	**Modern & Applied Optimization** Introduction to metaheuristics (GA, PSO); convex optimization in engineering; applications in communications and machine learning.	5
	Total Contact Hours (Lecture)	**28**

Text Books

1. S. S. Rao, Engineering Optimization: Theory and Practice, 4th ed., Wiley.
2. H. A. Taha, Operations Research: An Introduction, Pearson.

Reference Books

1. S. Boyd, L. Vandenberghe, Convex Optimization, Cambridge University Press.
2. E. K. P. Chong, S. H. Zak, An Introduction to Optimization, Wiley.

Industry Relevance

Optimization is central to resource allocation in wireless networks, VLSI placement/routing, control design, and the training of machine-learning models. These techniques are widely applied in telecom planning, semiconductor EDA, and operations research roles.

Open Educational Resources (OERs) & Open Repositories

- NPTEL: Optimization Techniques — https://nptel.ac.in/courses/111105100
- Boyd Convex Optimization (Stanford) lectures & book — https://web.stanford.edu/~boyd/cvxbook/
- MIT OCW 6.252 Nonlinear Programming — https://ocw.mit.edu/
- CVXPY documentation & examples — https://www.cvxpy.org/

Suggested Laboratory Exercises

1. Solve LP problems with the simplex method and SciPy/PuLP.
2. Transportation/assignment problem solutions.
3. Implement gradient descent and Newton's method.
4. Constrained optimization with KKT/penalty methods.
5. Convex optimization modelling with CVXPY.

Open-Source Simulators / Tools

Python (SciPy, PuLP, CVXPY) • GNU Octave • GLPK • Scilab

EC-L-05-C Digital Signal Processing

Course Code	EC-L-05-C	**Credits**	3
Category	PCC (Programme Core)	**Structure (L-T-P-S)**	2-0-2-2
Prerequisite	Signals and Systems	**Laboratory**	Mandatory

Covers discrete-time signal processing, transforms, and digital filter design with practical implementation.

Course Outcomes (COs)

On successful completion of this course, the student will be able to:

CO	Statement
CO1	Analyse discrete-time signals and systems in time and frequency.
CO2	Compute and apply the DFT and FFT.
CO3	Design IIR digital filters.
CO4	Design FIR digital filters.
CO5	Implement DSP algorithms and understand realization structures.

Detailed Syllabus (5 Modules)

Mod.	Content	Hours
M1	**Discrete-Time Systems** Discrete signals and systems; LTI systems; convolution; difference equations; z-transform and inverse; system stability.	5
M2	**DFT & FFT** DFT properties; circular convolution; linear filtering via DFT; radix-2 FFT algorithms (DIT/DIF); spectral analysis.	6
M3	**IIR Filter Design** Analog filter approximations (Butterworth, Chebyshev); impulse invariance; bilinear transformation; frequency transformations.	6
M4	**FIR Filter Design** Linear-phase FIR; windowing methods; frequency-sampling; Parks-McClellan (optimal) design; comparison with IIR.	6
M5	**Realization & Applications** Direct, cascade, parallel, lattice structures; finite word-length effects; multirate basics; DSP processors and applications.	5
	Total Contact Hours (Lecture)	**28**

Text Books

1. J. G. Proakis, D. G. Manolakis, Digital Signal Processing, 4th ed., Pearson.
2. A. V. Oppenheim, R. W. Schafer, Discrete-Time Signal Processing, Pearson.

Reference Books

1. S. K. Mitra, Digital Signal Processing: A Computer-Based Approach, McGraw-Hill.
2. E. C. Ifeachor, B. W. Jervis, Digital Signal Processing: A Practical Approach, Pearson.

Industry Relevance

DSP is core to audio/video, communications, biomedical, radar, and speech processing, and underpins software-defined radio and DSP-processor firmware in countless products.

Open Educational Resources (OERs) & Open Repositories

- NPTEL: Digital Signal Processing — https://nptel.ac.in/courses/108104100
- DSPRelated.com — https://www.dsprelated.com/
- MIT OCW 6.341 Discrete-Time Signal Processing — https://ocw.mit.edu/
- The Scientist and Engineer's Guide to DSP — https://www.dspguide.com/

Suggested Laboratory Exercises

1. z-transform and convolution in Python/Octave.
2. DFT/FFT and spectral analysis.
3. IIR filter design and verification.
4. FIR filter design with windows.
5. Real-time-style filtering of audio signals.
6. Implement a DSP block in SDR (GNU Radio).

Open-Source Simulators / Tools

Python (SciPy, NumPy) • GNU Octave • GNU Radio • pyFDA (filter design)

EC-L-06-C Communication System

Course Code	EC-L-06-C	**Credits**	3
Category	PCC (Programme Core)	**Structure (L-T-P-S)**	2-0-2-2
Prerequisite	Signals and Systems	**Laboratory**	Mandatory

Introduces analog and the foundations of digital communication: modulation, noise, and receiver performance.

Course Outcomes (COs)

On successful completion of this course, the student will be able to:

CO	Statement
CO1	Analyse amplitude and angle modulation schemes.
CO2	Evaluate noise performance of analog communication systems.
CO3	Apply sampling, quantization, and pulse modulation.
CO4	Explain the building blocks of a digital communication link.
CO5	Use simulation tools to model communication systems.

Detailed Syllabus (5 Modules)

Mod.	Content	Hours
M1	**Random Processes** Multiple random variables, Gaussian Random Variables, Random processes, Stationary processes, Statistical independence, Autocorrelation, Auto-covariance functions, Cross-Correlation, Ergodicity, Power spectral density, Relationship with autocorrelation, Random processes	7
M2	**Amplitude Modulation** AM, DSB-SC, SSB, VSB; modulators and demodulators; spectra and power; superheterodyne receiver. Frequency and phase modulation; narrowband and wideband FM; bandwidth (Carson's rule); FM generation and detection.	6
M3	**Noise** noise sources; SNR; noise in AM and FM receivers; figure of merit; pre-emphasis/de-emphasis.	5
M4	**Pulse Modulation** Sampling theorem; PAM, PWM, PPM; quantization; PCM; companding; delta and DPCM. ADM, A-law, line coding; multiplexing (TDM/FDM);	5
M5	**Digital Communication** Baseband transmission, digital-modulation overview; eye diagram. ASK, FSK, PSK, QPSK, QAM; signal-space representation; constellation diagrams; bandwidth efficiency	5
	Total Contact Hours (Lecture)	**28**

Text Books

1. S. Haykin, Communication Systems, 4th ed., Wiley.
2. B. P. Lathi, Z. Ding, Modern Digital and Analog Communication Systems, Oxford University Press.

Reference Books

1. T. S. Rappaport, Wireless Communications, Pearson.
2. J. G. Proakis, M. Salehi, Communication Systems Engineering, Pearson.

Industry Relevance

Analog/digital communication fundamentals underpin all telecom and broadcasting systems and are prerequisites for 4G/5G, satellite, and RF roles in companies such as Ericsson, Nokia, Qualcomm, and ISRO.

Open Educational Resources (OERs) & Open Repositories

- NPTEL: Principles of Communication Systems — https://nptel.ac.in/courses/108102096
- GNU Radio tutorials — https://wiki.gnuradio.org/index.php/Tutorials
- MIT OCW 6.450 Principles of Digital Communications — https://ocw.mit.edu/
- DSPRelated communication articles — https://www.dsprelated.com/

Suggested Laboratory Exercises

1. AM/FM modulation-demodulation simulation.
2. Noise effect and SNR measurement.
3. Sampling, quantization, and PCM.
4. Line coding and eye-diagram analysis.
5. GNU Radio flowgraph for a simple link.

Open-Source Simulators / Tools

GNU Radio • Python (NumPy, SciPy, CommPy) • GNU Octave • Scilab

CS-L-03-C Design and Analysis of Algorithms

Course Code	CS-L-03-C	**Credits**	3
Category	PCC (Programme Core)	**Structure (L-T-P-S)**	2-0-2-2
Prerequisite	Data Structures	**Laboratory**	Mandatory

Teaches systematic algorithm-design paradigms and complexity analysis for efficient problem solving.

Course Outcomes (COs)

On successful completion of this course, the student will be able to:

CO	Statement
CO1	Analyse algorithm time and space complexity using asymptotic notation.
CO2	Apply divide-and-conquer to design efficient algorithms.
CO3	Apply greedy and dynamic-programming techniques.
CO4	Solve graph problems with standard algorithms.
CO5	Recognize NP-hardness and apply approximation/backtracking.

Detailed Syllabus (5 Modules)

Mod.	Content	Hours
M1	**Foundations** Asymptotic notation; recurrence relations; master theorem; analysis of iterative and recursive algorithms.	6
M2	**Divide and Conquer** Merge sort; quick sort; binary search; matrix multiplication (Strassen); closest pair.	5
M3	**Greedy Methods** Activity selection; Huffman coding; minimum spanning trees (Kruskal, Prim); Dijkstra's shortest path.	5
M4	**Dynamic Programming** Principle of optimality; 0/1 knapsack; LCS; matrix-chain multiplication; Bellman-Ford; Floyd-Warshall.	6
M5	**Complexity & Intractability** Backtracking; branch and bound; P, NP, NP-complete; reductions; approximation algorithms.	6
	Total Contact Hours (Lecture)	**28**

Text Books

1. T. H. Cormen, C. E. Leiserson, R. L. Rivest, C. Stein, Introduction to Algorithms, 3rd ed., MIT Press.
2. E. Horowitz, S. Sahni, S. Rajasekaran, Fundamentals of Computer Algorithms, Universities Press.

Reference Books

1. J. Kleinberg, E. Tardos, Algorithm Design, Pearson.
2. S. Dasgupta, C. Papadimitriou, U. Vazirani, Algorithms, McGraw-Hill.

Industry Relevance

Algorithmic design and complexity analysis are foundational for EDA tools, network routing, optimization, and high-performance computing, and are central to technical hiring across the software and core-electronics industries.

Open Educational Resources (OERs) & Open Repositories

- MIT OCW 6.006 / 6.046 Algorithms — https://ocw.mit.edu/courses/6-046j-design-and-analysis-of-algorithms-spring-2015/
- NPTEL: Design and Analysis of Algorithms — https://nptel.ac.in/courses/106106131
- VisuAlgo — https://visualgo.net/
- CP-Algorithms — https://cp-algorithms.com/

Suggested Laboratory Exercises

1. Implement and benchmark sorting algorithms.
2. Divide-and-conquer problems (closest pair, etc.).
3. Greedy: MST and Dijkstra.
4. Dynamic programming: knapsack, LCS.
5. Backtracking: N-Queens / graph colouring.

Open-Source Simulators / Tools

GCC / Python • VS Code • VisuAlgo • Online Judges (Codeforces/LeetCode)

CS-L-04-C Object Oriented Programming

Course Code	CS-L-04-C	**Credits**	2
Category	PCC (Programme Core)	**Structure (L-T-P-S)**	1-0-2-2
Prerequisite	C Programming	**Laboratory**	Mandatory

Introduces object-oriented design and programming concepts using C++/Java for building modular, reusable software.

Course Outcomes (COs)

On successful completion of this course, the student will be able to:

CO	Statement
CO1	Apply OOP concepts: classes, objects, encapsulation, abstraction.
CO2	Implement inheritance and polymorphism.
CO3	Use operator/function overloading and templates.
CO4	Handle exceptions and perform file I/O.
CO5	Design small applications using OOP principles.

Detailed Syllabus (5 Modules)

Mod.	Content	Hours
M1	**OOP Basics** Procedural vs object-oriented paradigm; classes and objects; access specifiers; constructors and destructors; this pointer.	3
M2	**Inheritance** Types of inheritance; base/derived classes; virtual base class; access control in inheritance.	3
M3	**Polymorphism** Function and operator overloading; virtual functions; runtime polymorphism; abstract classes.	3
M4	**Templates & STL** Function and class templates; generic programming; introduction to the Standard Template Library (containers, iterators).	3
M5	**Exceptions & Files** Exception handling; streams and file I/O; introduction to design patterns.	2
	Total Contact Hours (Lecture)	**14**

Text Books

1. E. Balagurusamy, Object Oriented Programming with C++, McGraw-Hill.
2. B. Stroustrup, The C++ Programming Language, Addison-Wesley.

Reference Books

1. H. Schildt, Java: The Complete Reference, McGraw-Hill.
2. R. Lafore, Object-Oriented Programming in C++, Sams.

Industry Relevance

OOP underlies modern application, tool, and GUI development, and is used in EDA scripting, simulation frameworks, and embedded middleware. OOP fluency is expected for software-leaning roles in electronics and IT firms.

Open Educational Resources (OERs) & Open Repositories

- NPTEL: Programming in C++ — https://nptel.ac.in/courses/106105151
- cppreference.com — https://en.cppreference.com/
- learncpp.com — https://www.learncpp.com/
- GeeksforGeeks OOP — https://www.geeksforgeeks.org/object-oriented-programming-in-cpp/

Suggested Laboratory Exercises

1. Class/object design and constructors.
2. Inheritance hierarchies.
3. Operator and function overloading.
4. Templates and STL containers.
5. Exception handling and file I/O mini-project.

Open-Source Simulators / Tools

g++ / Clang • VS Code • Code::Blocks • OnlineGDB

EC-L-07-C Electronic Circuit-1

Course Code	EC-L-07-C	**Credits**	3
Category	PSC (Programme Specific Core)	**Structure (L-T-P-S)**	2-0-2-0
Prerequisite	Basic Electronics	**Laboratory**	Mandatory

Covers semiconductor devices and analog circuit design with diodes and transistors, the foundation of analog electronics.

Course Outcomes (COs)

On successful completion of this course, the student will be able to:

CO	Statement
CO1	Analyse diode circuits and rectifier/regulator behaviour.
CO2	Determine BJT/MOSFET biasing and operating point.
CO3	Analyse small-signal amplifier configurations.
CO4	Evaluate amplifier frequency response.
CO5	Design and simulate basic analog circuits.

Detailed Syllabus (5 Modules)

Mod.	Content	Hours
M1	**Diodes & Applications** PN junction; diode characteristics; clippers, clampers; half/full-wave rectifiers; filters; Zener regulators.	5
M2	**BJT Fundamentals** BJT structure and operation; characteristics; DC load line; biasing techniques; thermal stability.	6
M3	**BJT Amplifiers** Small-signal model; CE, CB, CC configurations; gain, input/output impedance; analysis.	5
M4	**FET & MOSFET** JFET and MOSFET operation and characteristics; biasing; common-source/drain amplifiers.	6
M5	**Frequency Response** Low- and high-frequency response; Miller effect; Bode plots; multistage amplifier basics.	6
	Total Contact Hours (Lecture)	**14**

Text Books

1. A. S. Sedra, K. C. Smith, Microelectronic Circuits, Oxford University Press.
2. R. L. Boylestad, L. Nashelsky, Electronic Devices and Circuit Theory, Pearson.

Reference Books

1. D. A. Neamen, Microelectronics: Circuit Analysis and Design, McGraw-Hill.
2. J. Millman, C. Halkias, Integrated Electronics, McGraw-Hill.

Industry Relevance

Analog circuit design skills are essential for power management, sensor front-ends, RF, and mixed-signal IC design, with strong demand in semiconductor and analog-IC companies (Texas Instruments, Analog Devices).

Open Educational Resources (OERs) & Open Repositories

- NPTEL: Analog Electronic Circuits — https://nptel.ac.in/courses/108105158
- All About Circuits — https://www.allaboutcircuits.com/
- Falstad Circuit Simulator — https://www.falstad.com/circuit/
- ngspice documentation — https://ngspice.sourceforge.io/docs.html

Suggested Laboratory Exercises

1. Diode characteristics and rectifier circuits.
2. BJT biasing and Q-point measurement.
3. CE amplifier gain and frequency response.
4. MOSFET common-source amplifier.
5. SPICE simulation of analog circuits.

Open-Source Simulators / Tools

ngspice / Qucs-S • KiCad • LTspice • Falstad • Multisim Live

EC-L-08-C Electromagnetic Field Theory

Course Code	EC-L-08-C	**Credits**	2
Category	PSC (Programme Specific Core)	**Structure (L-T-P-S)**	1-0-2-0
Prerequisite	Calculus, Vector Algebra	**Laboratory**	Faculty Choice

Builds understanding of static and time-varying electromagnetic fields and wave propagation, fundamental to antennas, RF, and microwave engineering.

Course Outcomes (COs)

On successful completion of this course, the student will be able to:

CO	Statement
CO1	Apply vector calculus to electrostatic and magnetostatic fields.
CO2	Use Maxwell's equations to analyse time-varying fields.
CO3	Derive and interpret the wave equation and Poynting theorem.
CO4	Analyse plane-wave propagation in different media.
CO5	Evaluate wave reflection and transmission at interfaces.

Detailed Syllabus (5 Modules)

Mod.	Content	Hours
M1	**Vector Analysis & Electrostatics** Coordinate systems; gradient, divergence, curl; Coulomb's and Gauss's laws; electric potential; capacitance.	3
M2	**Magnetostatics** Biot-Savart and Ampere's laws; magnetic flux density; magnetic forces; inductance; boundary conditions.	3
M3	**Maxwell's Equations** Faraday's law; displacement current; Maxwell's equations in integral and differential form; boundary conditions.	3
M4	**Wave Equation** Derivation of the wave equation in different coordinates; solutions; Poynting theorem and power flow.	3
M5	**Plane Waves** Uniform plane waves in lossless and lossy media; polarization; reflection and transmission at media interfaces.	2
	Total Contact Hours (Lecture)	**14**

Text Books

1. M. N. O. Sadiku, Elements of Electromagnetics, Oxford University Press.
2. W. H. Hayt, J. A. Buck, Engineering Electromagnetics, McGraw-Hill.

Reference Books

1. D. K. Cheng, Field and Wave Electromagnetics, Pearson.
2. J. D. Kraus, Electromagnetics, McGraw-Hill.

Industry Relevance

EM field theory is the basis of antenna, RF, microwave, EMC, and high-speed PCB design used in wireless, radar, aerospace, and defence sectors (e.g., ISRO, DRDO, RF/antenna design teams).

Open Educational Resources (OERs) & Open Repositories

- NPTEL: Electromagnetic Fields — https://nptel.ac.in/courses/108106073
- MIT OCW 8.02 Electricity and Magnetism — https://ocw.mit.edu/courses/8-02-physics-ii-electricity-and-magnetism-spring-2007/
- FEMM (Finite Element Method Magnetics) — https://www.femm.info/wiki/HomePage
- openEMS (open electromagnetic field solver) — https://www.openems.de/

Suggested Laboratory Exercises

1. Field visualization (E/H) using Python/Octave.
2. Capacitance/inductance computation for geometries.
3. Plane-wave propagation and polarization plots.
4. Reflection/transmission coefficient computation.
5. 2-D field simulation with FEMM/openEMS.

Open-Source Simulators / Tools

openEMS • FEMM • Python (Matplotlib, NumPy) • MEEP (FDTD)

DA-L-03-C Machine Learning Algorithms

Course Code	DA-L-03-C	**Credits**	3
Category	PCC (Programme Core)	**Structure (L-T-P-S)**	2-0-2-2
Prerequisite	Probability & Statistics, Introduction to AI & ML	**Laboratory**	Mandatory

Provides a rigorous treatment of supervised, unsupervised, and neural-network learning algorithms with hands-on implementation.

Course Outcomes (COs)

On successful completion of this course, the student will be able to:

CO	Statement
CO1	Formulate and evaluate supervised-learning models.
CO2	Apply regularization, model selection, and validation.
CO3	Implement unsupervised learning and dimensionality reduction.
CO4	Build and train basic neural networks.
CO5	Apply ML to signal/communication engineering problems.

Detailed Syllabus (5 Modules)

Mod.	Content	Hours
M1	**Supervised Learning I** Linear and logistic regression; gradient descent; bias-variance trade-off; regularization (ridge, lasso).	6
M2	**Supervised Learning II** Decision trees; ensemble methods (bagging, random forests, boosting); SVM and kernels; k-NN; Naïve Bayes.	6
M3	**Unsupervised Learning** k-means and hierarchical clustering; PCA; t-SNE; Gaussian mixture models; anomaly detection.	5
M4	**Neural Networks** Perceptron; multilayer networks; backpropagation; activation functions; introduction to deep learning and CNN.	6
M5	**Evaluation & Applications** Cross-validation; metrics; hyperparameter tuning; ML for communications/signal processing; responsible AI.	5
	Total Contact Hours (Lecture)	**28**

Text Books

1. C. M. Bishop, Pattern Recognition and Machine Learning, Springer.
2. A. Géron, Hands-On Machine Learning with Scikit-Learn, Keras and TensorFlow, O'Reilly.

Reference Books

1. T. Hastie, R. Tibshirani, J. Friedman, The Elements of Statistical Learning, Springer.
2. I. Goodfellow, Y. Bengio, A. Courville, Deep Learning, MIT Press.

Industry Relevance

ML algorithms power intelligent communication, edge-AI, predictive maintenance, and EDA optimization. Strong ML implementation skills are highly valued across telecom, semiconductor, and data-science roles.

Open Educational Resources (OERs) & Open Repositories

- Stanford CS229 Machine Learning — https://cs229.stanford.edu/
- NPTEL: Machine Learning — https://nptel.ac.in/courses/106106202
- fast.ai Practical Deep Learning — https://course.fast.ai/
- scikit-learn / TensorFlow tutorials — https://scikit-learn.org/stable/auto_examples/

Suggested Laboratory Exercises

1. Regression with regularization.
2. Classification with SVM, trees, and ensembles.
3. Clustering and PCA on real data.
4. Build and train a neural network (Keras/PyTorch).
5. ML applied to a signal-classification task.

Open-Source Simulators / Tools

Python (scikit-learn, TensorFlow, PyTorch, pandas) • Google Colab • Jupyter • Kaggle

EC-L-09-C Embedded Systems Design

Course Code	EC-L-09-C	**Credits**	3
Category	PCC (Programme Core)	**Structure (L-T-P-S)**	2-0-2-2
Prerequisite	Digital Electronics, C Programming	**Laboratory**	Mandatory

Covers microcontroller architecture, peripherals, interfacing, and real-time firmware design for embedded applications.

Course Outcomes (COs)

On successful completion of this course, the student will be able to:

CO	Statement
CO1	Explain embedded-system architecture and design metrics.
CO2	Program microcontrollers and configure on-chip peripherals.
CO3	Interface sensors, actuators, and communication peripherals.
CO4	Apply interrupts, timers, and real-time concepts.
CO5	Design and debug a complete embedded application.

Detailed Syllabus (5 Modules)

Mod.	Content	Hours
M1	**Embedded Fundamentals** Embedded-system characteristics and design metrics; microprocessor vs microcontroller; ARM Cortex-M architecture; memory and bus organization; toolchain.	6
M2	**I/O & Peripherals** GPIO; timers/counters; PWM; ADC/DAC; watchdog; clock and power management.	6
M3	**Interrupts & Communication** Interrupt handling and priorities; serial protocols - UART, SPI, I2C; CAN basics.	6
M4	**RTOS Concepts** Polling vs interrupt-driven design; tasks, scheduling, semaphores, queues; introduction to FreeRTOS.	5
M5	**System Design** Sensor/actuator interfacing; low-power design; debugging and testing; case studies and embedded product design.	5
	Total Contact Hours (Lecture)	**28**

Text Books

1. F. Vahid, T. Givargis, Embedded System Design: A Unified Hardware/Software Introduction, Wiley.
2. J. Yiu, The Definitive Guide to ARM Cortex-M3/M4 Processors, Elsevier.

Reference Books

1. R. Kamal, Embedded Systems: Architecture, Programming and Design, McGraw-Hill.
2. M. Mazidi et al., The STM32/ARM Microcontroller and Embedded Systems, Pearson.

Industry Relevance

Embedded design skills are in high demand in automotive (ADAS, EV), consumer electronics, medical devices, and industrial automation. Microcontroller firmware, RTOS, and peripheral interfacing are core to embedded-engineer roles.

Open Educational Resources (OERs) & Open Repositories

- NPTEL: Embedded Systems Design — https://nptel.ac.in/courses/108102169
- STM32 / ARM mbed documentation — https://os.mbed.com/docs/
- FreeRTOS documentation — https://www.freertos.org/Documentation/RTOS_book.html
- Wokwi (ESP32/STM32 simulator) — https://wokwi.com/

Suggested Laboratory Exercises

1. GPIO, timers, and PWM on an ARM Cortex-M board.
2. ADC-based sensor reading and display.
3. UART/SPI/I2C communication with peripherals.
4. Interrupt-driven event handling.
5. FreeRTOS multi-tasking example.
6. Mini embedded project (e.g., data logger).

Open-Source Simulators / Tools

STM32CubeIDE • Arduino IDE • PlatformIO • Wokwi • QEMU • FreeRTOS

EC-L-10-C Digital Communication I

Course Code	EC-L-10-C	**Credits**	3
Category	PCC (Programme Core)	**Structure (L-T-P-S)**	2-0-2-2
Prerequisite	Communication System, Probability	**Laboratory**	Mandatory

Covers digital transmission, baseband and passband modulation, and optimal receiver design over noisy channels.

Course Outcomes (COs)

On successful completion of this course, the student will be able to:

CO	Statement
CO1	Apply compression, and source coding.
CO2	Analyse baseband transmission and ISI.
CO3	Design and evaluate digital modulation schemes.
CO4	Apply optimal detection and compute error probabilities.
CO5	Apply channel-coding techniques.

Detailed Syllabus (5 Modules)

Mod.	Content	Hours
M1	**Source Coding & Digitization** source coding, Shannon Fano, Hoffman coding and information measures; entropy. Information theory	5
M2	**Baseband Transmission** pulse shaping; Nyquist criterion; inter-symbol interference; eye diagram; equalization basics.	6
M3	**Optimal Reception** Matched filter; correlator receiver; ML/MAP detection; probability of error for binary and M-ary schemes; AWGN channel.	6
M4	**Performance & Synchronization** BER performance comparison; carrier and symbol synchronization; introduction to spread spectrum.	5
M5	**Channel Coding** Linear block codes; cyclic codes; convolutional codes; Viterbi decoding; introduction to LDPC and turbo codes; coding gain.	8
	Total Contact Hours (Lecture)	**28**

Text Books

1. J. G. Proakis, M. Salehi, Digital Communications, 5th ed., McGraw-Hill.
2. S. Haykin, Digital Communication Systems, Wiley.

Reference Books

1. B. Sklar, Digital Communications: Fundamentals and Applications, Pearson.
2. B. P. Lathi, Z. Ding, Modern Digital and Analog Communication Systems, Oxford.

Industry Relevance

Digital communication is the foundation of all modern wireless and wireline systems (4G/5G/6G, Wi-Fi, satellite). Modulation, detection, and BER analysis skills are directly used in modem/PHY design at telecom and chipset companies.

Open Educational Resources (OERs) & Open Repositories

- NPTEL: Digital Communication — https://nptel.ac.in/courses/117101051
- MIT OCW 6.450 Principles of Digital Communications — https://ocw.mit.edu/
- CommPy (open communications library) — https://github.com/veeresht/CommPy
- GNU Radio digital tutorials — https://wiki.gnuradio.org/index.php/Tutorials

Suggested Laboratory Exercises

1. PCM and quantization-noise study.
2. Line coding and eye-diagram analysis.
3. ASK/FSK/PSK/QAM modulation and constellations.
4. Matched-filter receiver and BER vs SNR curves.
5. End-to-end link simulation in GNU Radio.

Open-Source Simulators / Tools

Python (NumPy, SciPy, CommPy) • GNU Radio • GNU Octave

EC-L-11-C Applied Antenna Engineering

Course Code	EC-L-11-C	**Credits**	3
Category	PCC (Programme Core)	**Structure (L-T-P-S)**	2-0-2-0
Prerequisite	Electromagnetic Field Theory	**Laboratory**	Mandatory

Covers antenna fundamentals, common antenna types, arrays, and measurement, with simulation-based design.

Course Outcomes (COs)

On successful completion of this course, the student will be able to:

CO	Statement
CO1	Define and compute antenna parameters.
CO2	Analyse wire and aperture antennas.
CO3	Design and analyse antenna arrays.
CO4	Select antennas for wireless and emerging applications.
CO5	Simulate and characterize antennas using EM tools.

Detailed Syllabus (5 Modules)

Mod.	Content	Hours
M1	**Antenna Fundamentals** Radiation mechanism; antenna parameters - gain, directivity, radiation pattern, beamwidth, polarization, impedance, bandwidth; Friis equation.	6
M2	**Wire Antennas** Infinitesimal and half-wave dipoles; monopole; loop antennas; effect of ground; baluns.	6
M3	**Aperture & Microstrip Antennas** Horn and reflector antennas; microstrip patch antennas; feeding techniques; design basics.	6
M4	**Antenna Arrays** Array factor; uniform linear arrays; broadside and end-fire; beam steering; introduction to phased arrays and MIMO antennas.	5
M5	**Applications & Measurement** Antennas for mobile/5G/IoT; mmWave and reconfigurable antennas; measurement setups; anechoic chamber basics.	5
	Total Contact Hours (Lecture)	**28**

Text Books

1. C. A. Balanis, Antenna Theory: Analysis and Design, Wiley.
2. J. D. Kraus, R. J. Marhefka, Antennas for All Applications, McGraw-Hill.

Reference Books

1. W. L. Stutzman, G. A. Thiele, Antenna Theory and Design, Wiley.
2. R. Garg et al., Microstrip Antenna Design Handbook, Artech House.

Industry Relevance

Antenna design is critical for mobile, 5G/mmWave, satellite, radar, and IoT devices. Simulation-driven antenna engineering is a specialized, well-paid skill in wireless, aerospace, and defence industries.

Open Educational Resources (OERs) & Open Repositories

- NPTEL: Antennas — https://nptel.ac.in/courses/108101092
- openEMS antenna tutorials — https://www.openems.de/index.php/Tutorial.html
- 4nec2 (open NEC-based antenna modeler) — https://www.qsl.net/4nec2/
- Antenna-Theory.com — https://www.antenna-theory.com/

Suggested Laboratory Exercises

1. Compute and plot dipole/monopole patterns.
2. Design a half-wave dipole and measure parameters.
3. Microstrip patch antenna design and simulation.
4. Linear array factor and beam steering.
5. EM simulation of a patch antenna (openEMS).

Open-Source Simulators / Tools

openEMS • 4nec2 • NEC2 • Python (Matplotlib) • Qucs

EC-L-12-C Electronics Circuit-2

Course Code	EC-L-12-C	**Credits**	3
Category	PCC (Programme Core)	**Structure (L-T-P-S)**	2-0-2-0
Prerequisite	Electronic Circuit-1	**Laboratory**	Mandatory

Covers feedback amplifiers, op-amps, oscillators, and analog ICs for advanced analog circuit design.

Course Outcomes (COs)

On successful completion of this course, the student will be able to:

CO	Statement
CO1	Analyse feedback amplifier topologies and stability.
CO2	Design op-amp-based linear and nonlinear circuits.
CO3	Analyse oscillators and waveform generators.
CO4	Design active filters and voltage regulators.
CO5	Simulate and test analog IC-based circuits.

Detailed Syllabus (5 Modules)

Mod.	Content	Hours
M1	**Feedback Amplifiers** Feedback concept; topologies; effect on gain, bandwidth, impedance; stability and compensation; Nyquist criterion.	6
M2	**Operational Amplifiers** Ideal/practical op-amp; inverting/non-inverting; summing, difference, integrator, differentiator; instrumentation amplifier.	6
M3	**Oscillators** Barkhausen criterion; RC (phase-shift, Wien-bridge), LC (Hartley, Colpitts), and crystal oscillators.	5
M4	**Active Filters & Comparators** Active low/high/band-pass filters; Butterworth/Chebyshev; comparators; Schmitt trigger; 555 timer; waveform generators.	6
M5	**Power & Regulators** Power amplifiers (Class A, B, AB, C); voltage regulators; linear and switching regulators; PLL basics.	5
	Total Contact Hours (Lecture)	**28**

Text Books

1. A. S. Sedra, K. C. Smith, Microelectronic Circuits, Oxford University Press.
2. R. A. Gayakwad, Op-Amps and Linear Integrated Circuits, Pearson.

Reference Books

1. D. A. Neamen, Microelectronics: Circuit Analysis and Design, McGraw-Hill.
2. S. Franco, Design with Operational Amplifiers and Analog ICs, McGraw-Hill.

Industry Relevance

Advanced analog design is essential for power management, signal conditioning, instrumentation, and mixed-signal ICs, with strong demand in analog/mixed-signal semiconductor companies.

Open Educational Resources (OERs) & Open Repositories

- NPTEL: Analog Circuits — https://nptel.ac.in/courses/108106182
- TI Analog Engineer's resources — https://www.ti.com/analog-circuit/op-amps.html
- LTspice / ngspice — https://ngspice.sourceforge.io/
- All About Circuits — https://www.allaboutcircuits.com/

Suggested Laboratory Exercises

1. Feedback amplifier gain/bandwidth measurement.
2. Op-amp linear circuits (amplifier, integrator).
3. Wien-bridge / phase-shift oscillator.
4. Active filter design and response.
5. 555-timer and Schmitt-trigger circuits.
6. SPICE simulation of analog IC circuits.

Open-Source Simulators / Tools

ngspice / LTspice • Qucs-S • KiCad • Falstad • Multisim Live

EC-L-13-C Network Analysis

Course Code	EC-L-13-C	**Credits**	2
Category	PCC (Programme Core)	**Structure (L-T-P-S)**	1-0-2-0
Prerequisite	Basic Circuits	**Laboratory**	Mandatory

Covers circuit-analysis theorems, transient/steady-state response, and two-port networks.

Course Outcomes (COs)

On successful completion of this course, the student will be able to:

CO	Statement
CO1	Apply network theorems to solve circuits.
CO2	Analyse transient response of RL, RC, RLC circuits.
CO3	Perform steady-state AC analysis using phasors.
CO4	Characterize two-port networks.
CO5	Apply Laplace transform to network analysis.

Detailed Syllabus (5 Modules)

Mod.	Content	Hours
M1	**Network Theorems** KVL/KCL; mesh and nodal analysis; superposition; Thevenin and Norton; maximum power transfer; reciprocity.	3
M2	**Transient Analysis** Initial conditions; first- and second-order circuits; natural and step response of RL, RC, RLC.	3
M3	**AC Steady-State** Phasors; impedance/admittance; AC power; resonance in series/parallel RLC; quality factor.	3
M4	**Two-Port Networks** Z, Y, h, and ABCD parameters; interconnection; image and characteristic impedance.	3
M5	**Laplace & Network Functions** Laplace-domain analysis; transfer functions; poles and zeros; network synthesis basics.	2
	Total Contact Hours (Lecture)	**14**

Text Books

1. M. E. Van Valkenburg, Network Analysis, Pearson.
2. C. K. Alexander, M. N. O. Sadiku, Fundamentals of Electric Circuits, McGraw-Hill.

Reference Books

1. W. H. Hayt, J. E. Kemmerly, Engineering Circuit Analysis, McGraw-Hill.
2. A. Chakrabarti, Circuit Theory, Dhanpat Rai.

Industry Relevance

Network analysis is fundamental to filter design, power electronics, RF matching, and PCB/SI analysis, and underlies nearly all hardware-design work.

Open Educational Resources (OERs) & Open Repositories

- NPTEL: Networks and Systems — https://nptel.ac.in/courses/108105159
- All About Circuits — https://www.allaboutcircuits.com/
- Falstad Circuit Simulator — https://www.falstad.com/circuit/
- ngspice — https://ngspice.sourceforge.io/

Suggested Laboratory Exercises

1. Verify Thevenin/Norton and superposition.
2. Transient response of RL/RC/RLC.
3. Resonance and Q-factor measurement.
4. Two-port parameter measurement.
5. Laplace-based response simulation.

Open-Source Simulators / Tools

ngspice • Falstad • LTspice • Python (SciPy)

EC-L-14-C Image Processing and Computer Vision

Course Code	EC-L-14-C	**Credits**	3
Category	PCC (Programme Core)	**Structure (L-T-P-S)**	2-0-2-0
Prerequisite	Signals and Systems; Problem Solving with C / Python programming	**Laboratory**	Mandatory

Provides a foundation in digital image formation, processing, and analysis, progressing to classical and deep-learning-based computer vision techniques for object detection, recognition, and scene understanding.

Course Outcomes (COs)

On successful completion of this course, the student will be able to:

CO	Statement
CO1	Explain digital image formation, sampling, quantization, and colour models.
CO2	Apply spatial- and frequency-domain techniques for image enhancement and filtering.
CO3	Perform image segmentation, edge detection, and morphological operations.
CO4	Extract and describe image features for matching, recognition, and tracking.
CO5	Design and evaluate CNN-based computer vision pipelines for classification and detection tasks.

Detailed Syllabus (5 Modules)

Mod.	Content	Hours
M1	**Digital Image Fundamentals** — Image formation and acquisition; sampling and quantization; pixel relationships; colour models (RGB, HSV, YCbCr); image file formats.	5
M2	**Image Enhancement & Filtering** — Spatial-domain enhancement (histogram equalization, smoothing, sharpening); frequency-domain processing (2D-DFT, filtering); noise models and denoising	6
M3	**Segmentation & Morphology** — Edge detection (Sobel, Canny); thresholding; region-based and watershed segmentation; morphological operations (erosion, dilation, opening, closing).	6
M4	**Feature Extraction & Matching** — Corner and blob detectors (Harris, SIFT, ORB); HOG features; feature matching and homography; image alignment and stitching.	6
M5	**Computer Vision with Deep Learning** — CNN architectures for vision; transfer learning; object detection (YOLO/SSD overview); semantic segmentation; introduction to vision transformers.	5
	Total Contact Hours (Lecture)	**28**

Text Books

1. R.C. Gonzalez, R.E. Woods, *Digital Image Processing*, 4th ed., Pearson.
2. R. Szeliski, *Computer Vision: Algorithms and Applications*, 2nd ed., Springer.

Reference Books

1. S. Prince, *Computer Vision: Models, Learning, and Inference*, Cambridge University Press.
2. A. Bovik, *Handbook of Image and Video Processing*, Academic Press.
3. I. Goodfellow, Y. Bengio, A. Courville, *Deep Learning*, MIT Press. (for CNN foundations)

Industry Relevance

Image processing and computer vision underpin autonomous vehicles, medical imaging diagnostics, industrial quality inspection, surveillance and biometrics, AR/VR, and satellite/remote-sensing analytics — core technologies across the semiconductor, automotive, healthcare, and AI industries.

Open Educational Resources (OERs) & Open Repositories

- NPTEL: Digital Image Processing — https://nptel.ac.in/courses/117105079
- Stanford CS231n: Convolutional Neural Networks for Visual Recognition — http://cs231n.stanford.edu
- OpenCV Documentation & Tutorials — https://docs.opencv.org

Suggested Laboratory Exercises

1. Implement histogram equalization and spatial filtering on grayscale/colour images.
2. Perform frequency-domain filtering using 2D-FFT.
3. Implement edge detection and morphological operations for object isolation.
4. Extract and match SIFT/ORB features between image pairs; build a simple panorama stitcher.
5. Train a CNN for image classification on a standard dataset (e.g., CIFAR-10).
6. Build an object-detection pipeline using a pretrained model (e.g., YOLO) on custom images.

Open-Source Simulators / Tools

Python (OpenCV, scikit-image, NumPy) • PyTorch / TensorFlow • MATLAB Image Processing Toolbox • Jupyter

EC-L-16-C Microelectronics

Course Code	EC-L-16-C	**Credits**	3
Category	PCC (Programme Core)	**Structure (L-T-P-S)**	2-0-2-2
Prerequisite	Electronic Circuits, Semiconductor Devices	**Laboratory**	Mandatory

Covers the physics, fabrication, and modelling of semiconductor devices that form the building blocks of modern integrated circuits.

Course Outcomes (COs)

On successful completion of this course, the student will be able to:

CO	Statement
CO1	Explain carrier transport and the physics of semiconductor junctions.
CO2	Analyse the operation and characteristics of BJTs and MOSFETs.
CO3	Model devices for circuit simulation using small- and large-signal models.
CO4	Describe the major steps of CMOS IC fabrication.
CO5	Relate device-level non-idealities to circuit and chip performance.

Detailed Syllabus (5 Modules)

Mod.	Content	Hours
M1	**Semiconductor Physics** Energy bands; intrinsic and extrinsic semiconductors; carrier statistics; drift and diffusion; continuity equation; generation-recombination.	6
M2	**PN Junction & BJT** PN-junction electrostatics; I-V characteristics; capacitances; BJT operation, regions, Ebers-Moll model; small-signal model.	6
M3	**MOS Devices** MOS capacitor; threshold voltage; MOSFET I-V characteristics; body effect; channel-length modulation; sub-threshold conduction; short-channel effects.	6
M4	**Device Modelling** SPICE device models; large- and small-signal models; temperature dependence; noise in devices; scaling and its implications.	5
M5	**IC Fabrication** Crystal growth; oxidation; photolithography; diffusion and ion implantation; etching; metallisation; CMOS process flow; packaging.	5
	Total Contact Hours (Lecture)	**28**

Text Books

1. B. G. Streetman, S. K. Banerjee, Solid State Electronic Devices, Pearson.
2. S. M. Sze, K. K. Ng, Physics of Semiconductor Devices, Wiley.

Reference Books

1. D. A. Neamen, Semiconductor Physics and Devices, McGraw-Hill.
2. R. F. Pierret, Semiconductor Device Fundamentals, Pearson.

Industry Relevance

Microelectronics is the foundation of the semiconductor industry. Device physics, MOSFET modelling, and CMOS fabrication knowledge underpins roles in chip design, process engineering, and device characterisation at companies such as Intel, TSMC, Samsung, and Indian semiconductor-mission fabs.

Open Educational Resources (OERs) & Open Repositories

- NPTEL: Semiconductor Devices and Circuits — https://nptel.ac.in/courses/108108122
- MIT OCW 6.012 Microelectronic Devices and Circuits — https://ocw.mit.edu/
- nanoHUB device simulation tools — https://nanohub.org/
- Georgia Tech / online SPICE references — https://ngspice.sourceforge.io/docs.html

Suggested Laboratory Exercises

1. Diode and BJT characteristic extraction in SPICE.
2. MOSFET I-V and transfer-characteristic simulation.
3. Extract threshold voltage and trans-conductance.
4. Temperature dependence of device parameters.
5. CMOS inverter DC characteristics from device models.

Open-Source Simulators / Tools

Ngspice • Qucs-S • nanoHUB online tools • KLayout (layout viewing)

EC-L-17-C Digital Communication II

Course Code	EC-L-17-C	**Credits**	3
Category	PCC (Programme Core)	**Structure (L-T-P-S)**	2-0-2-2
Prerequisite	Digital Communication I, Probability & Statistics	**Laboratory**	Mandatory

Advances digital communication theory into channel coding, equalisation, spread spectrum, and modern multi-carrier and multi-antenna techniques used in 4G/5G systems.

Course Outcomes (COs)

On successful completion of this course, the student will be able to:

CO	Statement
CO1	Analyse performance of digital modulation over fading channels.
CO2	.
CO3	Design equalisers to combat inter-symbol interference.
CO4	Explain spread-spectrum and multi-carrier (OFDM) systems.
CO5	Describe principles of MIMO and modern multiple-access schemes.

Detailed Syllabus (5 Modules)

Mod.	Content	Hours
M1	**Wireless Channel** Path loss, shadowing, Multipath fading channel, Delay spread, Doppler spread, Rayleigh fading channel, Rician fading channel	6
M2	**Channel Models & Detection** AWGN and fading channels; flat vs frequency-selective fading; diversity techniques; optimum receivers; BER analysis over fading.	5
M3	**Equalisation** ISI and Nyquist criterion; linear equalisers - ZF and MMSE; decision-feedback equalisation; adaptive equalisation.	5
M4	**Spread Spectrum & Multi-Carrier** Direct-sequence and frequency-hopping spread spectrum; PN sequences; CDMA; OFDM principles; cyclic prefix; PAPR.	6
M5	**MIMO & Multiple Access** MIMO channel capacity; spatial multiplexing and diversity; space-time coding basics; OFDMA, SC-FDMA; introduction to NOMA.	5
	Total Contact Hours (Lecture)	**28**

Text Books

1. J. G. Proakis, M. Salehi, Digital Communications, McGraw-Hill.
2. B. Sklar, Digital Communications: Fundamentals and Applications, Pearson.

Reference Books

1. A. Goldsmith, Wireless Communications, Cambridge University Press.
2. D. Tse, P. Viswanath, Fundamentals of Wireless Communication, Cambridge University Press.

Industry Relevance

Channel coding, OFDM, and MIMO are core technologies of 4G LTE, 5G NR, and Wi-Fi. These skills directly map to physical-layer design, modem development, and wireless-systems engineering at telecom and chipset firms such as Qualcomm, Ericsson, Nokia, and MediaTek.

Open Educational Resources (OERs) & Open Repositories

- NPTEL: Principles of Digital Communications — https://nptel.ac.in/courses/108102117
- MIT OCW 6.450 Principles of Digital Communications — https://ocw.mit.edu/
- GNU Radio tutorials — https://wiki.gnuradio.org/
- CommPy (open-source comm library) — https://github.com/veeresht/CommPy

Suggested Laboratory Exercises

1. BER simulation of modulation over fading channels.
2. Convolutional encoder and Viterbi decoder.
3. ZF/MMSE equaliser implementation.
4. OFDM transceiver with cyclic prefix.
5. Simple 2x2 MIMO spatial-multiplexing simulation.

Open-Source Simulators / Tools

Python (NumPy, SciPy, CommPy) • GNU Octave • GNU Radio • Scilab

EC-L-18-C Control Systems

Course Code	EC-L-18-C	**Credits**	3
Category	PCC (Programme Core)	**Structure (L-T-P-S)**	2-0-2-0
Prerequisite	Signals and Systems	**Laboratory**	Mandatory

Introduces analysis and design of linear feedback control systems in time and frequency domains, with applications to electronic and robotic systems.

Course Outcomes (COs)

On successful completion of this course, the student will be able to:

CO	Statement
CO1	Model physical systems using transfer functions and state-space.
CO2	Analyse time-domain transient and steady-state response.
CO3	Assess stability using Routh-Hurwitz, root locus, and frequency methods.
CO4	Design compensators to meet performance specifications.
CO5	Apply state-space methods for controllability and observability.

Detailed Syllabus (5 Modules)

Mod.	Content	Hours
M1	**System Modelling** Open- and closed-loop systems; transfer functions; block diagrams and signal-flow graphs; modelling of electrical/mechanical systems.	6
M2	**Time-Domain Analysis** First- and second-order responses; transient specifications; steady-state error; error constants; sensitivity.	6
M3	**Stability & Root Locus** Concept of stability; Routh-Hurwitz criterion; root-locus construction and analysis; effect of poles and zeros.	6
M4	**Frequency-Domain Analysis** Bode plots; Nyquist criterion; gain and phase margins; lead, lag, and lead-lag compensator design.	5
M5	**State-Space Methods** State-space representation; solution of state equations; controllability and observability; state-feedback and pole placement.	5
	Total Contact Hours (Lecture)	**28**

Text Books

1. K. Ogata, Modern Control Engineering, Pearson.
2. N. S. Nise, Control Systems Engineering, Wiley.

Reference Books

1. B. C. Kuo, F. Golnaraghi, Automatic Control Systems, Wiley.
2. R. C. Dorf, R. H. Bishop, Modern Control Systems, Pearson.

Industry Relevance

Control systems are essential in robotics, UAVs, automotive systems, process automation, and power electronics. Feedback design, stability analysis, and state-space control are core competencies for control, mechatronics, and autonomous-systems engineers.

Open Educational Resources (OERs) & Open Repositories

- NPTEL: Control Engineering — https://nptel.ac.in/courses/108106098
- MIT OCW 6.302 Feedback Systems — https://ocw.mit.edu/
- Python Control Systems Library — https://python-control.readthedocs.io/
- Scilab/Xcos control resources — https://www.scilab.org/

Suggested Laboratory Exercises

1. Time response of first/second-order systems.
2. Root-locus design and analysis.
3. Bode and Nyquist stability evaluation.
4. Lead-lag compensator design.
5. State-feedback pole-placement simulation.

Open-Source Simulators / Tools

Python (python-control, NumPy) • GNU Octave • Scilab/Xcos

EC-L-19-C VLSI Design and IC Technology

Course Code	EC-L-19-C	**Credits**	3
Category	PCC (Programme Core)	**Structure (L-T-P-S)**	2-0-2-2
Prerequisite	Digital Electronics, Microelectronics	**Laboratory**	Mandatory

Covers CMOS digital VLSI design from logic to layout, including combinational/sequential design, timing, power, and the standard IC design flow.

Course Outcomes (COs)

On successful completion of this course, the student will be able to:

CO	Statement
CO1	Design CMOS logic gates and analyse their characteristics.
CO2	Apply MOSFET sizing for performance and power.
CO3	Design combinational and sequential CMOS circuits.
CO4	Analyse delay, power, and noise in digital VLSI circuits.
CO5	Follow the standard-cell ASIC/IC design and layout flow.

Detailed Syllabus (5 Modules)

Mod.	Content	Hours
M1	**CMOS Fundamentals** MOS switch model; CMOS inverter - VTC, noise margins, switching; static and dynamic power; transistor sizing.	6
M2	**Combinational Logic** Static CMOS logic; complex gates; ratioed and pass-transistor logic; transmission gates; dynamic logic styles.	6
M3	**Sequential Logic** Latches and flip-flops; timing parameters; clocking strategies; setup/hold; clock distribution and skew.	5
M4	**Delay, Power & Interconnect** RC delay models; logical effort; power dissipation components; low-power techniques; interconnect modelling and scaling.	6
M5	**VLSI Design Flow** ASIC design flow; HDL synthesis; floorplanning, placement, routing; design rules; layout, DRC/LVS; introduction to FPGA flow.	5
	Total Contact Hours (Lecture)	**28**

Text Books

1. N. Weste, D. Harris, CMOS VLSI Design: A Circuits and Systems Perspective, Pearson.
2. J. Rabaey, A. Chandrakasan, B. Nikolic, Digital Integrated Circuits, Pearson.

Reference Books

1. S. M. Kang, Y. Leblebici, CMOS Digital Integrated Circuits, McGraw-Hill.
2. W. Wolf, Modern VLSI Design, Pearson.

Industry Relevance

VLSI design is the core of the semiconductor and chip-design industry. CMOS logic design, timing/power analysis, and the RTL-to-GDSII flow are directly relevant to ASIC/SoC design roles at firms driving India's semiconductor mission and global EDA/chip companies.

Open Educational Resources (OERs) & Open Repositories

- NPTEL: Digital VLSI Design — https://nptel.ac.in/courses/106106087
- MIT OCW 6.374 Analysis and Design of Digital ICs — https://ocw.mit.edu/
- OpenLane / open-source RTL-to-GDSII flow — https://github.com/The-OpenROAD-Project/OpenLane
- VLSI System Design (vsdflow) resources — https://www.vlsisystemdesign.com/

Suggested Laboratory Exercises

1. CMOS inverter VTC and delay simulation.
2. Static CMOS gate design and verification.
3. Sequential element (flip-flop) timing analysis.
4. RTL synthesis with Yosys.
5. Standard-cell layout and DRC using Magic/KLayout.

Open-Source Simulators / Tools

Ngspice • Yosys • Magic VLSI • KLayout • OpenLane (Sky130)

EC-L-20-C FPGA based System Design

Course Code	EC-L-20-C	**Credits**	2
Category	PCC (Programme Core)	**Structure (L-T-P-S)**	1-0-2-0
Prerequisite	Digital Electronics using Verilog	**Laboratory**	Mandatory

Covers FPGA architecture, advanced HDL design, and the synthesis-to-implementation flow for reconfigurable systems.

Course Outcomes (COs)

On successful completion of this course, the student will be able to:

CO	Statement
CO1	Explain FPGA architecture and design flow.
CO2	Write synthesizable Verilog/VHDL for FPGAs.
CO3	Design FSMs and datapaths for FPGA implementation.
CO4	Apply timing constraints and analyse timing.
CO5	Implement and verify a system on an FPGA/soft-core.

Detailed Syllabus (5 Modules)

Mod.	Content	Hours
M1	**FPGA Architecture** PLD to FPGA evolution; CLBs, LUTs, flip-flops, routing, I/O blocks, DSP slices, block RAM; SRAM vs flash FPGAs.	3
M2	**HDL Design for FPGA** Synthesizable RTL coding styles; combinational vs sequential; parameterized modules; coding for inference of RAM/DSP.	3
M3	**Design Flow** Synthesis, mapping, placement, routing; bitstream generation; simulation and testbenches; IP integration.	3
M4	**Timing & Constraints** Setup/hold; clock domains and CDC; timing constraints (SDC); static timing analysis; pipelining.	3
M5	**System Design** Soft-core processors (e.g., RISC-V/PicoRV32); AXI interfacing; case studies - UART, SPI, signal-processing block.	2
	Total Contact Hours (Lecture)	**14**

Text Books

1. P. P. Chu, FPGA Prototyping by Verilog Examples, Wiley.
2. S. Kilts, Advanced FPGA Design, Wiley.

Reference Books

1. W. Wolf, FPGA-Based System Design, Pearson.
2. Vendor user guides (AMD-Xilinx Vivado, Intel Quartus).

Industry Relevance

FPGA design is widely used in telecom, defence, high-frequency trading, and prototyping of ASICs. RTL coding, timing closure, and FPGA implementation are core to hardware-design and verification careers.

Open Educational Resources (OERs) & Open Repositories

- NPTEL: Hardware Modeling using Verilog — https://nptel.ac.in/courses/106105165
- FPGA4Fun — https://www.fpga4fun.com/
- Project IceStorm (open FPGA toolchain) — https://clifford.at/icestorm
- EDA Playground — https://www.edaplayground.com/

Suggested Laboratory Exercises

1. Combinational and sequential designs on FPGA.
2. FSM-based controller implementation.
3. UART/SPI module on FPGA.
4. Timing constraints and STA exercise.
5. Soft-core / signal-processing block deployment.

Open-Source Simulators / Tools

AMD-Xilinx Vivado (free) • Intel Quartus Prime Lite • Yosys + nextpnr (open) • Icarus Verilog + GTKWave

EC-L-22-C Microwave, mmWave & RFIC for Emerging Technologies

Course Code	EC-L-22-C	**Credits**	3
Category	PCC (Programme Core)	**Structure (L-T-P-S)**	2-0-2-2
Prerequisite	Electromagnetic Field Theory, Electronic Circuits	**Laboratory**	Mandatory

Covers microwave and millimetre-wave engineering, passive/active RF components, and RFIC design relevant to 5G/6G and emerging wireless technologies.

Course Outcomes (COs)

On successful completion of this course, the student will be able to:

CO	Statement
CO1	Analyse transmission lines and microwave networks using S-parameters.
CO2	Design passive microwave components and matching networks.
CO3	Explain operation of RF amplifiers, mixers, and oscillators.
CO4	Describe mmWave propagation and front-end design challenges.
CO5	Apply RFIC concepts to emerging 5G/6G transceiver design.

Detailed Syllabus (5 Modules)

Mod.	Content	Hours
M1	**Microwave Fundamentals** Transmission-line theory; Smith chart; impedance matching; S-parameters; microwave network analysis.	6
M2	**Passive Components** Waveguides; couplers; power dividers; filters; resonators; ferrite devices; planar (microstrip) components.	6
M3	**Active RF Circuits** RF amplifier design; low-noise amplifiers; power amplifiers; mixers; oscillators; noise figure and linearity.	6
M4	**mmWave & Antennas** mmWave propagation; path loss and atmospheric effects; phased arrays; beamforming; antenna arrays for 5G/6G.	5
M5	**RFIC & Emerging Systems** RFIC design considerations; transceiver architectures; CMOS/SiGe RF; integration challenges; trends toward 6G and THz.	5
	Total Contact Hours (Lecture)	**28**

Text Books

1. D. M. Pozar, Microwave Engineering, Wiley.
2. T. H. Lee, The Design of CMOS Radio-Frequency Integrated Circuits, Cambridge University Press.

Reference Books

1. B. Razavi, RF Microelectronics, Pearson.
2. R. Ludwig, G. Bogdanov, RF Circuit Design: Theory and Applications, Pearson.

Industry Relevance

Microwave, mmWave, and RFIC expertise is central to 5G/6G base-stations, satellite links, radar, and RF front-ends. These skills map to RF design, antenna engineering, and transceiver roles at telecom, defence, and semiconductor companies, and connect strongly with IIIT-NR's 6G research.

Open Educational Resources (OERs) & Open Repositories

- NPTEL: Microwave Engineering — https://nptel.ac.in/courses/108101112
- MIT OCW RF/microwave resources — https://ocw.mit.edu/
- openEMS electromagnetic field solver — https://www.openems.de/
- scikit-rf (RF/microwave in Python) — https://scikit-rf.org/

Suggested Laboratory Exercises

1. Smith-chart impedance matching with scikit-rf.
2. S-parameter analysis of a two-port network.
3. Microstrip line and filter simulation.
4. LNA noise-figure and gain analysis.
5. Phased-array beam-pattern simulation.

Open-Source Simulators / Tools

scikit-rf (Python) • openEMS • Qucs-S • Ngspice

EC-L-23-C Testing and Verification of VLSI Circuits

Course Code	EC-L-23-C	**Credits**	2
Category	PCC (Programme Core)	**Structure (L-T-P-S)**	1-0-2-0
Prerequisite	VLSI Design and IC Technology	**Laboratory**	Mandatory

Covers fault modelling, test generation, design-for-test, and functional verification methods essential to the modern VLSI/SoC design flow.

Course Outcomes (COs)

On successful completion of this course, the student will be able to:

CO	Statement
CO1	Explain fault models and the need for VLSI testing.
CO2	Generate test patterns for combinational and sequential circuits.
CO3	Apply design-for-testability (DFT) techniques.
CO4	Describe built-in self-test (BIST) and boundary-scan methods.
CO5	Apply functional-verification methodology to digital designs.

Detailed Syllabus (5 Modules)

Mod.	Content	Hours
M1	**Testing Fundamentals** Need for testing; faults vs errors vs defects; fault models - stuck-at, bridging, delay; fault coverage; yield.	5
M2	**Test Generation** Fault simulation; combinational ATPG - D-algorithm, PODEM; sequential test generation; testability measures (SCOAP).	5
M3	**Design for Testability** Ad-hoc DFT; scan design; full and partial scan; test-point insertion; test-cost considerations.	5
M4	**BIST & Boundary Scan** Built-in self-test architectures; LFSR-based pattern generation and signature analysis; boundary scan (JTAG / IEEE 1149.1).	5
M5	**Verification Methodology** Simulation-based verification; testbenches; assertions; code/functional coverage; introduction to UVM concepts.	4
	Total Contact Hours (Lecture)	**24**

Text Books

1. M. Bushnell, V. Agrawal, Essentials of Electronic Testing for Digital, Memory and Mixed-Signal VLSI Circuits, Springer.
2. M. Abramovici, M. Breuer, A. Friedman, Digital Systems Testing and Testable Design, Wiley-IEEE.

Reference Books

1. L.-T. Wang, C.-W. Wu, X. Wen, VLSI Test Principles and Architectures, Morgan Kaufmann.
2. C. Spear, G. Tumbush, SystemVerilog for Verification, Springer.

Industry Relevance

Test and verification engineers are critical in every semiconductor company, as silicon cannot ship without thorough DFT and verification. Scan, BIST, ATPG, and verification skills map directly to DFT and design-verification (DV) roles in ASIC/SoC teams.

Open Educational Resources (OERs) & Open Repositories

- NPTEL: VLSI Testing and Verification — https://nptel.ac.in/courses/106105148
- cocotb (Python-based HDL verification) — https://www.cocotb.org/
- Verilator open-source simulator — https://www.veripool.org/verilator/
- EDA Playground (SystemVerilog/UVM) — https://www.edaplayground.com/

Suggested Laboratory Exercises

1. Stuck-at fault simulation on a combinational circuit.
2. ATPG pattern generation exercise.
3. Scan-chain insertion and verification.
4. LFSR-based BIST pattern generation.
5. Testbench with assertions and coverage using cocotb.

Open-Source Simulators / Tools

Icarus Verilog + GTKWave • Verilator • cocotb • Yosys

7. Professional Electives (Programme Electives)

The programme offers Professional/Programme Elective courses (PE-L-xx-E, 3 credits each) that allow students to build depth in one of three specialisation tracks aligned with the department's strengths and current industry demand. Students choose Programme Electives across Semesters 6 and 7 as per the scheme. A student may concentrate in a single track or combine courses across tracks. The lists below are indicative; the department may add or revise courses from time to time.

Track A — Communications Systems & Networks

Aligned with IIIT-NR's research in wireless communications, 6G, OTFS, and UAV-assisted networks.

Sl.	Course Code	Course Title	Credits
1	PE-L-01-E	Wireless and Mobile Communication	3
2	PE-L-02-E	5G/6G Networks and Architecture	3
3	PE-L-03-E	Information Theory and Coding	3
4	PE-L-04-E	Optical Fibre Communication	3
5	PE-L-05-E	Satellite and Space Communication	3
6	PE-L-06-E	Wireless Sensor Networks and IoT Communication	3
7	PE-L-07-E	Software-Defined Radio and Cognitive Radio	3
8	PE-L-08-E	Machine Learning for Wireless Communication	3

Track B — Embedded Systems & IoT

Focuses on real-time, connected, and intelligent embedded platforms for industrial and consumer applications.

Sl.	Course Code	Course Title	Credits
1	PE-L-09-E	Real-Time Operating Systems	3
2	PE-L-10-E	IoT Architecture and Edge Computing	3
3	PE-L-11-E	Embedded Linux and Device Drivers	3
4	PE-L-12-E	Automotive Electronics and CAN Systems	3
5	PE-L-13-E	ARM-based SoC Design	3
6	PE-L-14-E	TinyML and Edge-AI for Embedded Systems	3
7	PE-L-15-E	Sensor Systems and Signal Conditioning	3
8	PE-L-16-E	Cyber-Physical Systems	3

Track C — VLSI & Semiconductor Design

Builds depth in analog, digital, and physical IC design, supporting India's semiconductor-mission talent needs.

Sl.	Course Code	Course Title	Credits
1	PE-L-17-E	CMOS Analog IC Design	3
2	PE-L-18-E	Digital VLSI Design (Advanced)	3
3	PE-L-19-E	Low-Power VLSI Design	3
4	PE-L-20-E	Physical Design and Design Automation	3

Sl.	Course Code	Course Title	Credits
5	PE-L-21-E	SoC Design and Verification	3
6	PE-L-22-E	Mixed-Signal IC Design	3
7	PE-L-23-E	FPGA-based Hardware Acceleration	3
8	PE-L-24-E	Hardware Security and Trust	3

8. Open Electives (Suggested)

Open Electives (OE-L-xx-E, 3 credits each) are taken in Semesters 6 and 7 and allow ECE students to broaden their competencies beyond the core discipline. The following are suggested Open Electives grouped by domain. Courses are drawn from sister departments (CSE/DSAI), the management group, and humanities & social sciences. The actual basket offered each term is published by the institute.

Domain I — Computer Science / Data Science & AI

Sl.	Course Code	Course Title	Credits
1	OE-L-01-E	Cloud Computing	3
2	OE-L-02-E	Cyber Security and Network Security	3
3	OE-L-03-E	Computer Networks	3
4	OE-L-04-E	Deep Learning	3
5	OE-L-05-E	Data Mining and Analytics	3
6	OE-L-06-E	Blockchain Technology	3
7	OE-L-07-E	Computer Vision	3
8	OE-L-08-E	Natural Language Processing	3

Domain II — Management

Sl.	Course Code	Course Title	Credits
1	OE-L-09-E	Project Management	3
2	OE-L-10-E	Financial Management	3
3	OE-L-11-E	Marketing Management	3
4	OE-L-12-E	Organizational Behaviour	3
5	OE-L-13-E	Operations and Supply-Chain Management	3
6	OE-L-14-E	Technology and Innovation Management	3

Domain III — Humanities & Social Sciences

Sl.	Course Code	Course Title	Credits
1	OE-L-15-E	Technical Writing and Communication	3
2	OE-L-16-E	Introduction to Psychology	3
3	OE-L-17-E	Principles of Economics	3
4	OE-L-18-E	Philosophy of Science and Technology	3
5	OE-L-19-E	Sociology of Technology	3
6	OE-L-20-E	Science, Technology and Society	3

Note: Multidisciplinary Minor (MDM) streams are administered separately under the institute's minor framework and are intentionally excluded from this syllabus book, as per scope.

HS-L-08-C Introduction to Finance

Course Code	HS-L-08-C	**Credits**	1
Category	HSMC (AEC)	**Structure (L-T-P-S)**	1-0-0-0
Prerequisite	NIL	**Laboratory**	No

Introduces students to the fundamental concepts of personal and corporate finance, building the financial literacy needed to make informed decisions about savings, investment, credit, and basic business finance.

Course Outcomes (COs)

On successful completion of this course, the student will be able to:

CO	Statement
CO1	Explain core financial concepts, including the time value of money and interest computation.
CO2	Compare savings, investment, and insurance instruments based on risk and return.
CO3	Apply budgeting and financial-planning principles to personal financial decisions.
CO4	Interpret basic corporate financial statements and key financial ratios.

Detailed Syllabus (5 Modules)

Mod.	Content	Hours
M1	**Foundations of Finance** — Overview of the financial system and institutions; time value of money; simple and compound interest; present value and future value; inflation and purchasing power.	5
M2	**Personal Finance & Investment** — Savings and budgeting; banking and digital payment systems; insurance basics; introduction to mutual funds and the stock market; risk-return trade-off; retirement and tax planning basics.	5
M3	**Corporate Finance Basics** — Reading financial statements (balance sheet, P&L, cash flow); key financial ratios; sources of business finance (equity, debt); basics of capital budgeting.	4
	Total Contact Hours (Lecture)	**14**

Text Books

1. Prasanna Chandra, *Fundamentals of Financial Management*, McGraw-Hill.
2. Z. Bodie, A. Kane, A. Marcus, *Investments*, McGraw-Hill.

Reference Books

1. M.Y. Khan, P.K. Jain, *Financial Management*, McGraw-Hill.
2. Benjamin Graham, *The Intelligent Investor*, HarperBusiness.

Industry Relevance

Financial literacy is essential for engineers managing project budgets, evaluating ESOPs/stock compensation, and navigating startup finance and personal wealth planning over a technical career.

Open Educational Resources (OERs) & Open Repositories

- NPTEL: Financial Management — https://nptel.ac.in/courses/110105031
- SEBI Investor Education — https://investor.sebi.gov.in
- Khan Academy: Personal Finance — https://www.khanacademy.org/college-careers-more/personal-finance

Suggested Learning Activities

- Prepare a personal monthly budget and savings plan.
- Compute present/future value for a sample loan or investment.
- Analyze the balance sheet and P&L of a listed company (case study).

NEP-2020 Based B.Tech - ECE- UG curriculum

(With effect from July 2025)

Total Credit : 160

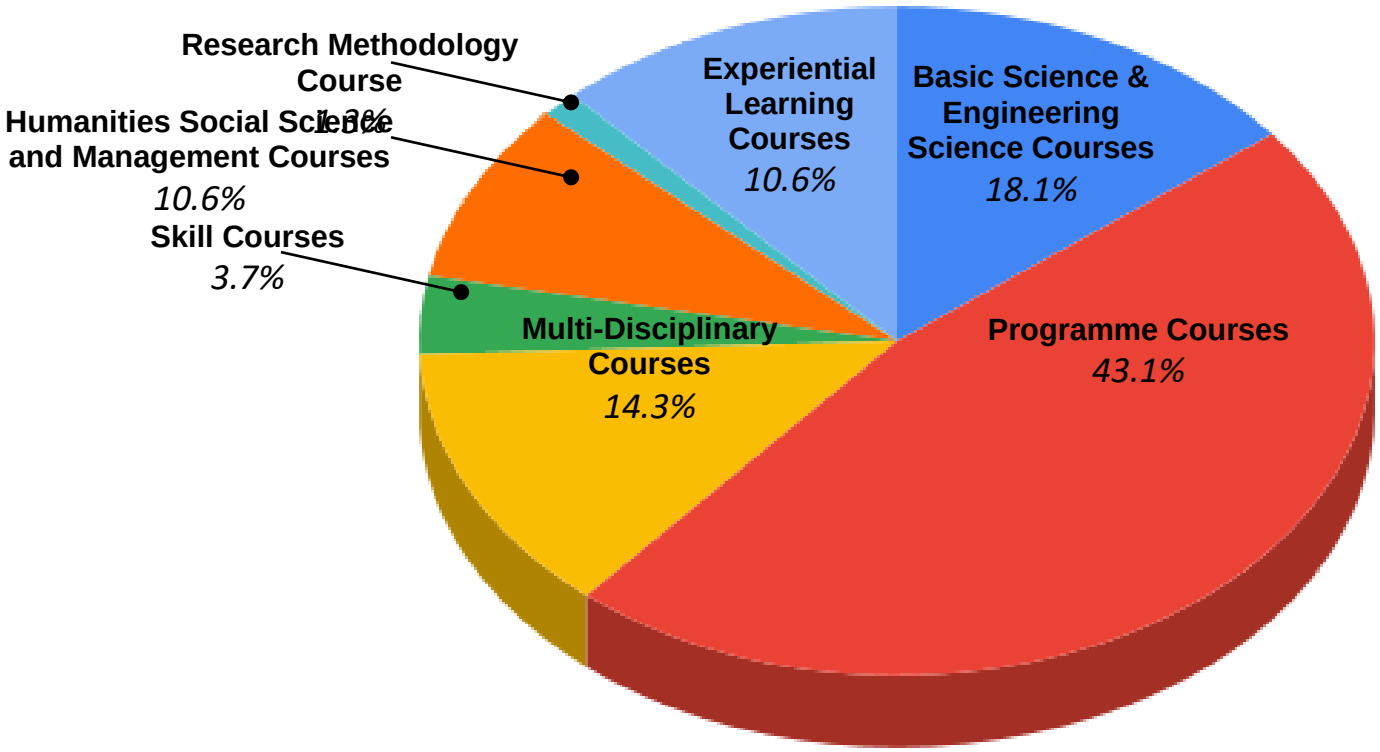
Lecture-L: 1 Credit = 1 hr/ week — Tutorial-T: 1 Credit = 1hr/week

Practical-P: 1 Credit = 2 hr/week — Scholarly Engagement Hours-S: 0 Credit= 2 hr/week

Multidisciplinary B.Tech - CSE Program: Sem-wise Credit Dist.- One Major, One Minor

Semester		I	II	III	IV	V	VI	VII	VIII	Total Credits	
Basic Science Course	BSC/ESC	06	06	03	--	--	--	--	--	**15**	**29**
Engineering Science Course		06	08		--	--	--	--	--	**14**	
Programme Core Course (PCC)	Programme Courses	--	--	15	19	15	05	–	--	**54**	**69**
Programme Elective Course (PEC)		--	--	--	--	–	09	06	--	**15**	
Open/School Elective (OE/SE) other than particular program	Multi-disciplinary	--	--	--	--	–	03	03	--	**06**	**18**
Multidisciplinary Minor (MD M)		--	--	03	03	03	03	03	--	**15**	
Vocational and Skill Enhancement Course (VSEC)	Skill Courses	02	04	--	–	--		--	--	**06**	**06**
Ability Enhancement Course (AEC-01)	Humanities Social Science and Management	02	02	--	--	--	--	--	--	**04**	**17**
Indian Language (AEC-02)		–	--	–	--	--	01	--	--	**01**	
Entrepreneurship/Economics/ Management Courses		02	--	--	–	02	--	–	--	**04**	
Indian Knowledge System (IKS)		–	–	02		--	--	02	--	**04**	
Value Education Course (VEC)		03	01	–	–	--	–	--	--	**04**	
Research Methodology (RM)	RM	--	--		--	02	--	–	--	**02**	**02**
Internship	Experiential Learning Courses	--	--		--	–	--	02			
Project		--	--		02	02	02	02	06	**16**	**17**

Community Engagement Activity (CEA)/Field Project		--	---	--	01	--	--	--	--	**01**	
Co-curricular & Extracurricular Activities (CCA)	Liberal Learning Course	–	–	--	--	--	02	--	--	**02**	**2**
Total Credits (One Major, One Minor degree)		**21**	**21**	**23**	**25**	**24**	**25**	**18**	**06**	**160**	**160**



Multidisciplinary Minor Streams

1. Cyber Security and Digital Forensics

Sl. No.	Name of the Course	Credits
1	Fundamentals of Cyber security	3
2	System Security	3
3	Cyber Threat Intelligence	3
4	Malware Analysis and Intrusion Detection Systems	3
5	Cyber Audit and Compliance	3

2. Business Informatics

Sl. No.	Name of the Course	Credits
1	Business Process Management	3
2	Introduction to FinTech	3
3	AI Based CRM	3
4	Digital Transformation using Process Mining	3
5	Strategic Supply Chain Management	3

3. Digital Humanities

Sl. No.	Name of the Course	Credits
1	Introduction to Digital Humanities	3
2	Digitalization of Heritage & Cultural Data	3
3	Digital Storytelling & Visualization	3
4	Culture Informatics: AI & Society	3
5	Spatial Humanities & GIS for Digital Mapping	3

4. Digital Mobility: Robotics and Unmanned Autonomous Systems

Sl. No.	Name of the Course	Credits
1	Fundamentals of Unmanned Autonomous System(UAS)	3
2	UAS Systems and Drone Technology	3
3	Autonomous Navigation and Motion Planning	3
4	UAS Data Analytics and Aerial Mapping	3
5	AI for Autonomous Systems	3

NEP Compliant Curriculum: HSM Courses to be included in UG Programs

Semester	Course category	Course Title	L-T-P-S	Credits	Course sub-category (*)
I	HSMC-01	International Language Competency	1-0-2-0	02	AEC
I	HSMC-02	Entrepreneurship and Design Thinking	1-0-0-0	01	-
I	HSMC-03	Human Values and Professional Ethics	2-0-0-0	02	VEC
I	HSMC-04	Constitution of India	1-0-0-0	01	VEC
II	HSMC-05	Communication & Discourse Strategies	2-0-0-0	02	AEC
II	HSMC-06	Environmental Studies	0-0-0-2	02	VEC
III	HSMC-07	Course on IKS in collaboration with Liberal Arts University	2-0-0-0	02	IKS
V	HSMC-08	Introduction to Finance	1-0-0-0	01	AEC
VI	HSMC-09	Indian/Foreign Language (One of the Following) Regional Language/German/Japanese	2-0-0-0	02	AEC
VII	HSMC-10	(One of the following) a) *History of Indian Civilizations* b) *Kautilya's Arthashastra* c) *Vedic Mathematics* d) *Vedic Corpus* e) *Wisdom from the Ages* f) *Panini's Grammar*	2-0-0-0	02	IKS
IV / VI	HSMC-11 / 12	Community Engagement Activities (NSS/ISR) / Extra and Co-curricular Activities	1 2	03	CEA CCA
		Total:	**13-0-2-2**	**20**	

- **AEC: Ability Enhancement Course**
- **IKS: Indian Knowledge System**
- **VEC: Value Education Course**
- **CEA: Community Engagement Activity**

Rubric Frameworks for Evaluation of Community Engagement Activities (NSS/ISR) / Extra and Co-Curricular Activities

Semester V

A. NSS (Hours of Service): NSS Coordinator will be responsible for providing the grade

Grade	A+	A	B+	B	C+	C	F
Hours of service	≥ 240	**200-239**	**180-199**	**140-179**	**100-139**	**60-99**	**<60**

B. ISR (30 Activities in Two years): ISR Coordinator will be responsible for providing the grade

Grade	A+	A	B+	B	C+	C	F
Present in % of activities	≥ 90%	**80%-89%**	**70%-79%**	**60%-69%**	**50%-59%**	**40%-49%**	**<40%**

a. Students can attend any one activity to earn the grade.
b. If a student wants to increase their grade, they need to participate in another activity also. Then the final grade will be decided by combining the grades of the two activities (NSS and ISR). For example, If a student scores an A in NSS and a C in ISR, his final grade will be A+. In general, students will be given one grade above the maximum grade of two activities, provided both grades are C and above.
c. For an F grade, the student has to do all ISR activities in the next semester.

Semester VI

Co-curricular & Extra-curricular Activities

Students' participation in the institute activities up to IV sem

a. Different clubs: technical and non-technical clubs; the coordinator of clubs will provide the names of students with attendance in the number of sessions and achievements
b. Sports activities: participation in institute sports activities and sports competition: The SAC president will be responsible for providing the participant's name
c. Attend IEEE activities and institute talk: The IEEE Coordinator will be responsible for providing the name
d. Volunteers in different technical and non-technical events. The faculty coordinator of particular activities will be responsible for providing the participant's name.

All activities will have equal weightage.

Total score (%)=$25\%\left(\frac{Total\ attendance}{Total\ number\ of\ session}+\frac{no\ times\ \ participation}{no\ of\ sport\ events}+\frac{No\ of\ talk\ attended}{Total\ number\ of\ talk}+\frac{No\ of\ event\ voluntered}{Total\ number\ of\ event}\right)$

Grade	A+	A	B+	B	C+	C	F
Total score (%)	≥ 90%	**80%-89%**	**70%-79%**	**60%-69%**	**50%-59%**	**40%-49%**	**<40%**